

RESEARCH ARTICLE

The Impact of News Sentiment on the Bitcoin Price via Machine Learning and Deep Learning-Based NLP Models

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Received: 27 April 2025 | **Revised:** 21 September 2025 | **Accepted:** 13 November 2025

Keywords: Bitcoin price | deep learning | machine learning | news sentiment analysis | NLP

ABSTRACT

This paper employs deep learning and machine learning-based NLP models to investigate the impact of the news sentiment on the Bitcoin price. The lagged Bitcoin variables, news indicators, macroeconomic, and financial factors were taken into account to explain the importance of news sentiment on the Bitcoin price. Moreover, FinBERT-based sentiment scores and semantic features extracted from over 650,000 financial news headlines were integrated with financial and macroeconomic variables. The importance scores of the investigation showed that Bitcoin was largely explained by its lagged price movements, which suggests the speculative nature of the cryptocurrency. However, the investigation also revealed that Bitcoin was significantly influenced by the news sentiment score. In other words, the paper indicates that the movements in the Bitcoin price can be predominantly explained by the news sentiment. Advanced hybrid models (all ML and DL models with the addition of variables obtained with the FinBERT model) were optimized using Optuna and RandomizedSearchCV. The FinBERT-LSTM model achieved the best prediction accuracy. Nevertheless, the main findings indicated that the response of the Bitcoin price to negative news was much stronger than to positive and neutral news. This finding suggests that the asymmetric relationship between the Bitcoin price and news sentiment was evident. GARCH-based volatility and what-if scenario analyses further demonstrated that negative sentiment leads to sharper fluctuations in the Bitcoin price. The paper provides important implications for policymakers, portfolio managers, investors, and academics.

JEL Classification: C30, C50

1 | Introduction

Globalization and rapid technological advancement have fundamentally transformed financial markets, paving the way for the emergence of next-generation financial instruments. Among these, Bitcoin has become one of the most prominent and widely debated innovations. As a decentralized digital asset with high volatility, Bitcoin challenges traditional perceptions of investment and payment systems. The accessibility of Bitcoin through advanced technological infrastructures has intensified the interest of policymakers, portfolio managers, investors, and scholars in understanding its role and implications in modern financial

markets. In light of these developments, this study aims to examine the determinants of Bitcoin price movements by employing cutting-edge artificial intelligence methods. Specifically, it utilizes hybrid models that integrate natural language processing (NLP) techniques with machine learning (ML) and deep learning (DL) algorithms. By leveraging FinBERT (Financial Bidirectional Encoder Representations from Transformers), a model fine-tuned for financial text analysis, the study assesses the influence of positive, negative, and neutral news on Bitcoin price dynamics. In addition, macroeconomic and financial indicators are incorporated to evaluate the predictive performance and robustness of the models.

Bitcoin's increasing appeal in global markets is underscored by its decentralized nature and extreme price volatility. This volatility, a defining characteristic of the asset, presents both lucrative opportunities and significant risks for market participants (Noviandy et al. 2023; Jesika et al. 2023). The speculative trading environment, coupled with Bitcoin's relatively low market capitalization compared to traditional assets such as equities or bonds, renders it particularly sensitive to large buy-sell transactions and external stimuli (Sahi et al. 2023; Zhang et al. 2024). Factors such as regulatory announcements and macroeconomic indicators further exacerbate its price instability. For instance, policy decisions and governmental statements concerning the regulation or endorsement of cryptocurrencies have historically triggered sharp reversals in the Bitcoin price (Lian 2022; Zhu 2023). Moreover, media coverage significantly shapes investor sentiment, often leading to abrupt price shifts in response to prominent news events (Coulter 2022; Kurnaman and Rizal 2023). Given these characteristics, understanding Bitcoin's pricing mechanisms necessitates an interdisciplinary approach that combines financial theory with computational intelligence. Accurate forecasting of Bitcoin's price movements is crucial not only for enhancing investor decision-making but also for equipping policymakers and economists with data-driven tools to manage systemic risks. The intricate interplay among speculative behavior, sentiment propagation through media, and broader macroeconomic signals shapes the unpredictable nature of Bitcoin's valuation. Consequently, evaluating these dimensions is critical for assessing Bitcoin's viability as an investment vehicle and its broader implications for financial stability.

This study seeks to answer several pivotal research questions: (i) To what extent do macroeconomic and financial variables, in conjunction with sentiment-laden news, influence the Bitcoin price? (ii) Is there evidence of asymmetric effects, particularly with respect to the differential impact of positive versus negative news? (iii) How can these insights inform policy and investment strategies? Additionally, the study investigates whether combining sentiment analysis features derived from FinBERT with conventional financial indicators improves predictive accuracy, and which hybrid architecture demonstrates the best overall performance. Furthermore, it explores how sentiment scores, categorized as positive, negative, or neutral, relate to price volatility and directional trends.

Conventional forecasting techniques, such as fundamental and technical analysis, often fall short when applied to the Bitcoin market due to its distinct structural and behavioral dynamics. While some technical indicators—like moving average convergence divergence (MACD) and various oscillators—offer limited utility, their effectiveness is frequently undermined by the rapid and erratic sentiment shifts typical of cryptocurrency trading (Khedr et al. 2021; Terentiev et al. 2021; Gyamerah 2021). The inability of these classical tools to adapt to non-stationary and nonlinear market conditions has prompted growing interest in artificial intelligence approaches. In particular, social media signals, macroeconomic developments, and geopolitical events often drive Bitcoin price movements in ways that defy rational market expectations (Băroiu and Băra 2024). This complexity has spurred calls for the adoption of machine learning frameworks, which are better suited to capturing high-dimensional,

nonlinear relationships and adapting to changing market conditions (Nasarudin et al. 2022; Xiang 2024). Among these, DL models—especially the long short-term memory (LSTM) networks—have demonstrated superior capabilities in modeling temporal dependencies and price fluctuations in financial time series, outperforming traditional approaches such as ARIMA (Shen et al. 2021; Sidiq and Nurzaman 2024). The hybridization of technical indicators with advanced learning algorithms represents a promising direction for financial forecasting, blending the explanatory power of classical metrics with the adaptability of AI models (Fousekis and Grigoriadis 2021).

Parallel to this evolution, sentiment analysis has emerged as a vital tool in capturing the emotional tone and interpretive content of financial discourse. Breaking news, tweets by influential actors, and broader social media dynamics have demonstrably immediate impacts on cryptocurrency markets (Girsang and Stanley 2023; Wang et al. 2024). Accordingly, NLP-based models—particularly those trained on domain-specific corpora—offer significant advantages in forecasting. FinBERT, designed specifically for financial texts, has proven particularly effective in extracting nuanced sentiment features that enhance the predictive performance of financial models (Mardjo and Choksuchat 2022; Belcastro et al. 2023).

In this study, both ML and DL techniques were employed to evaluate the efficacy of FinBERT-driven sentiment features. ML methods included extreme gradient boosting (XGBoost), random forest (RF), light gradient boosting machine (LightGBM), and support vector regression (SVR), while DL models comprised LSTM, gated recurrent unit (GRU), and convolutional neural network (CNN). These approaches were utilized to develop hybrid models that integrate quantitative indicators with textual sentiment data. The FinBERT-LSTM combination yielded the highest forecasting performance, indicating that Bitcoin price fluctuations are strongly influenced not only by lagged price values but also by the sentiment scores extracted from news media. Furthermore, to ensure linguistic diversity and reduce regional biases, the study incorporated news articles in 10 different languages during the sentiment extraction process. The findings confirmed an asymmetric relationship: negative news tends to exert a stronger and more immediate influence on price declines than positive or neutral news does on upward movements.

The remainder of this paper is structured as follows. Section 2 presents a comprehensive theoretical framework through an extensive review of relevant literature. Section 3 details the dataset, variable construction, and methodology. Section 4 discusses the empirical findings, including comparative model evaluations, scenario simulations, and volatility analyses. Section 5 provides a detailed discussion of the findings. Section 6 concludes the study by summarizing the key results, outlining both theoretical and practical implications, and suggesting directions for future research.

2 | Previous Research and the Current Work

In this section, the existing academic literature on Bitcoin price forecasting will be reviewed in detail under three main

headings. First, traditional econometric models that are widely used to explain Bitcoin price movements and studies that examine the impact of financial and macroeconomic factors on the Bitcoin price will be presented. We then discuss the recent literature on ML and DL models used to forecast the Bitcoin price. Finally, we review studies that examine the impact of sentiment analysis from financial news and social media sources on Bitcoin price movements and market volatility. At the end of the literature review, we will clarify the theoretical framework of this research by identifying the strengths and weaknesses of existing studies and highlighting key gaps in the literature that need to be filled.

2.1 | Predicting the Bitcoin Price Using Traditional Models and Financial Indicators

Much of the research on Bitcoin price forecasting emphasizes the use of traditional econometric models and financial indicators. These approaches generally examine the relationship between Bitcoin price movements and key economic variables in financial markets, employing classical time series models such as ARIMA, Generalized Autoregressive Conditional Variance (GARCH), and VAR (Berger and Koubová 2024; Kurnaman and Rizal 2023; Shen et al. 2021; Bouri et al. 2023). GARCH models, in particular, are frequently preferred due to their effectiveness in capturing the high volatility inherent in cryptocurrency markets. For instance, Kurnaman and Rizal (2023) demonstrated that Bitcoin price volatility is significantly influenced by trading volume and return rates using asymmetric GARCH models. Similarly, Shen et al. (2021) compared GARCH with recurrent neural networks (RNN) for volatility forecasting and noted that while traditional models offer advantages in modeling volatility structures, deep learning approaches often yield superior predictive accuracy. In addition to volatility modeling, several studies have investigated the link between the Bitcoin price and macroeconomic fundamentals. Caliciotti et al. (2024), employing both classical regression and machine learning techniques, found that indicators such as the stock-to-flow ratio, inflation rates, and interest rates are particularly influential in shaping long-term Bitcoin price cycles. Complementarily, Gyamerah (2021) enhanced the forecasting capacity of traditional models by integrating high-frequency price data with technical indicators and decomposition techniques like Variational Mode Decomposition. Nevertheless, the limitations of traditional econometric methods are also well-documented. Berger and Koubová (2024) argued that such models often fail to reflect the nonlinear and rapidly shifting dynamics of the Bitcoin price, especially in the short term. Similarly, Bouri et al. (2023) explored the relationship between sectoral volatility in the U.S. stock market and Bitcoin, concluding that while traditional models are useful for understanding macro-level interactions, they are insufficient for capturing the speculative and reactive nature of cryptocurrency markets in real time. In conclusion, although traditional time series and econometric models continue to play a vital role in explaining long-term structural patterns and macroeconomic dependencies, they fall short in accurately forecasting the short-term, speculative, and highly volatile movements of the Bitcoin market. This shortcoming

has led scholars to explore more comprehensive, adaptive, and hybrid modeling frameworks.

2.2 | Bitcoin Price Prediction by Machine Learning and Deep Learning Models

The extreme volatility and non-stationary nature of the Bitcoin price have challenged the capabilities of traditional financial models and directed research towards ML and DL approaches. These methods offer enhanced forecasting performance by capturing nonlinear dependencies inherent in financial markets (Tripathy et al. 2024; Alnami et al. 2025). Within ML, studies have frequently compared the predictive power of algorithms such as RF, SVR, XGB, and gradient boosting (GB). For instance, Basher and Sadorsky (2022) benchmarked ML techniques—including boosted regression trees (BRT) and support vector machines (SVM)—for forecasting cryptocurrency prices using high-dimensional macro-financial predictors. Their results showed that ensemble-based models like BRT significantly outperform traditional econometric frameworks, particularly under multi-step forecasting schemes. They also underscored the predictive relevance of macroeconomic indicators such as interest rates, inflation, and commodity prices, which align with the current study's emphasis on macro-financial integration. Similarly, Samson (2024) found that ML models including RF and GB offer superior accuracy in predicting Bitcoin and other cryptocurrency prices compared to traditional methods. Hussein and Abdulazeez (2024) expanded this comparison to include DL models such as LSTM and GRU, concluding that DL methods generally deliver better results than ML algorithms in price prediction tasks. Among DL approaches, LSTM, GRU, and CNN are particularly effective in learning both short- and long-term dependencies in sequential data. Tripathy et al. (2024) and Lamothe-Fernández et al. (2020) showed that LSTM and GRU outperform classical methods like ARIMA, especially during periods of high market volatility. Xiang (2024) further demonstrated the superiority of DL by developing a hybrid LSTM framework that successfully captured the complex behavior of cryptocurrency markets.

Despite their advantages, AI-based models present challenges such as overfitting, interpretability, and data imbalance. Chen (2023) noted that DL models (LSTM and GRU) are prone to overfitting on large datasets unless optimized with careful regularization. Qureshi et al. (2025) addressed this limitation by integrating synthetic minority over-sampling technique (SMOTE) to improve prediction robustness in imbalanced datasets. Overall, ML and DL models have been widely recognized in the literature for outperforming traditional statistical and econometric models in forecasting the Bitcoin price (Tsai et al. 2025). However, their success depends heavily on proper hyperparameter tuning, model optimization, and data preprocessing. These challenges highlight the necessity of hybrid and adaptive forecasting frameworks that integrate multiple data sources and modeling strategies.

Table 1 provides a comparative overview of selected ML and DL studies, summarizing their feature sets, contributions, and methodological limitations.

TABLE 1 | Comparison of machine learning and deep learning models for Bitcoin price prediction.

Study	Model(s) used	Feature types used	Main contribution	Limitations noted
Basher and Sadowsky (2022)	BRT, SVM	Macro-financial predictors	BRT outperforms others in multi-step forecasts	Limited handling of text sentiment
Chen (2023)	LSTM, GRU, RF	Price data	Identifies overfitting risk in DL models	No hybrid integration
Samson (2024)	RF, GB	Technical indicators, past prices	ML > traditional econometrics	Lack of macroeconomic integration
Hussein and Abdulazeez (2024)	RF, SVM, LSTM, GRU	Price, volume, technical indicators	DL outperforms ML overall	Overfitting risk, poor explainability
Tripathy et al. (2024)	LSTM, GRU	Financial time series	DL excels in volatility phases	High training cost
Xiang (2024)	Hybrid LSTM	Cryptocurrency data	Captures nonlinearity better than ARIMA	No macro or sentiment input
Qureshi et al. (2025)	ML + SMOTE	Imbalanced datasets	Improves prediction via data balancing	ML only, no text or macro features

2.3 | Social Media and News Based Sentiment Analysis and Hybrid Approaches

Recent years have witnessed a surge in studies integrating social media and financial news sentiment into Bitcoin price prediction models. This line of research rests on the premise that investor sentiment—shaped by public discourse and media narratives—significantly influences cryptocurrency price dynamics (Coulter 2022; Bouri et al. 2023). Textual data from platforms such as Twitter, Reddit, and financial news outlets are typically analyzed using NLP techniques to quantify market sentiment and incorporate it into forecasting models (Amirshahi and Lahmiri 2023).

Advanced transformer-based models like FinBERT, BERT, and GPT have proven especially effective in extracting financial sentiment. Liu et al. (2024) demonstrated that news-induced sentiment derived from these models plays a decisive role in anticipating market movements. Supporting this, Bâra and Oprea (2024) combined VADER and BERT sentiment scores with technical indicators within an LSTM framework. Their model not only captured negative sentiment shocks with high sensitivity but also benefited from the use of domain-specific sentiment features. Similarly, Loginova et al. (2024) utilized aspect-based sentiment analysis on Reddit and Bitcointalk data to achieve strong short-term prediction accuracy.

A recurring theme in the literature is the effectiveness of hybrid architectures that fuse sentiment inputs with other feature sets. For example, Patel et al. (2024) combined Twitter sentiment with a bidirectional LSTM model to predict hourly Bitcoin price. Jung et al. (2023) adopted a similar strategy using XGB with sentiment indicators from Reddit and LexisNexis, reporting substantial accuracy gains.

Several studies have found that sentiment-enhanced models are particularly resilient during extreme market conditions. Bâra et al. (2024) introduced a comprehensive meta-model that integrates sentiment, financial, and energy-related variables (e.g., carbon emissions, oil, and electricity prices) using SQL-based feature generation and stacking ensemble techniques. Their design, which emphasizes multidimensional inputs, aligns closely with the macroeconomic-integrated framework of the present study. Likewise, Rateb et al. (2024) showed that CNN-LSTM and SVM hybrids leveraging Google Trends and social media data provided robust performance during the Russia–Ukraine war. Jahanbin and Chahooki (2025) further advanced the field by demonstrating that sentiment extracted from influencer tweets, when processed through DistilBERT and BiGRU with attention, significantly improved trend prediction. However, the predictive efficacy of sentiment-based models still hinges on the quality of the NLP tools and source data. Mardjo and Chokschat (2022) highlighted that traditional sentiment tools like VADER yield lower accuracy in financial contexts compared to finance-specific models. Expanding on this, Girsang and Stanley (2023) showed that integrating FinBERT into LSTM and GRU networks enhances predictive accuracy for altcoins like Ethereum and Solana.

In summary, sentiment analysis has evolved into a central component of modern Bitcoin forecasting, especially when integrated into hybrid architectures with DL and ML components. Nevertheless, the reliability of such systems depends on rigorous NLP methodology, broad social data coverage, and effective integration with economic–financial indicators. These observations indicate a clear need for more sophisticated, multi-layered forecasting frameworks. Table 2 presents a comparative review of hybrid sentiment-based forecasting

TABLE 2 | Comparison of sentiment-based hybrid models in Bitcoin forecasting.

Study	NLP model(s)	Integrated features	Hybrid architecture	Key outcome
Mardjo and Choksuchat (2022)	VADER	Tweets	Random forest	Inferior to FinBERT in financial context
Girsang and Stanley (2023)	FinBERT	Ethereum and Solana financial texts	LSTM, GRU	FinBERT improves accuracy in altcoins
Jung et al. (2023)	Sentiment indices	Reddit, LexisNexis, technicals	XGB	Sentiment improves accuracy significantly
Bâra and Oprea (2024)	VADER, BERT	Sentiment + technical indicators	LSTM	LSTM captures negative shocks effectively
Bâra et al. (2024)	BERT	Energy prices + financial + sentiment	Meta-model with stacking	Improved performance across market regimes
Rateb et al. (2024)	Lexicon-based	Social media, Google Trends	CNN-LSTM, SVM	Stronger under extreme events
Liu et al. (2024)	FinBERT, BERT, GPT	Financial news texts	Standalone transformer models	FinBERT yields best results on sentiment extraction
Loginova et al. (2024)	Aspect-based sentiment analysis	Reddit, Bitcointalk forums	JST and TS-LDA	High accuracy for short-term prediction
Patel et al. (2024)	Lexicon-based	Twitter sentiment	BiLSTM	High hourly prediction performance
Jahanbin and Chahooki (2025)	DistilBERT	Influencer tweets	BiGRU + attention	Effective for trend forecasting

models, grouped by NLP method and architectural design. This format helps contextualize the evolution of sentiment integration in Bitcoin forecasting.

2.4 | The Current Work and Research Gap

Recent studies on Bitcoin price forecasting have approached the problem from varying methodological angles, each

contributing to a distinct aspect of understanding market dynamics. Traditional econometric models such as ARIMA, VAR, and GARCH remain relevant, particularly for capturing long-term trends and their association with macroeconomic indicators (Kurnaman and Rizal 2023; Berger and Koubová 2024). However, their limited capacity to model the high-frequency volatility and nonlinear behaviors that characterize Bitcoin markets has increasingly led scholars to explore more flexible and adaptive approaches (Gyamerah 2021; Shen et al. 2021).

In response to these limitations, the literature has witnessed a growing interest in AI-based models, especially DL techniques such as LSTM, CNN, and GRU, which are better equipped to handle complex and nonlinear relationships in financial time series (Tripathy et al. 2024; Xiang 2024). Despite their empirical success in improving forecasting accuracy, these models often rely exclusively on financial or technical indicators and overlook qualitative inputs such as investor sentiment and market psychology (Chen 2023; Qureshi et al. 2025). Additionally, common challenges such as data quality concerns, overfitting risks, and hyperparameter sensitivity continue to affect their robustness. To address this shortcoming, recent efforts have explored sentiment analysis approaches, leveraging news articles and social media content to incorporate investor sentiment into prediction models (Jahanbin and Chahooki 2025; Patel et al. 2024). Although these approaches enrich forecasting frameworks by integrating behavioral dimensions of market activity, many studies fall short of combining sentiment data with financial and macroeconomic indicators in a cohesive structure. Instead, they tend to model these components in isolation, resulting in fragmented and partial explanations of price dynamics (Loginova et al. 2024; Amirshahi and Lahmiri 2023). Moreover, a majority of the NLP methods employed in these studies rely on general-purpose language models, with limited use of domain-specific tools such as FinBERT that are explicitly trained on financial text (Liu et al. 2024; Girsang and Stanley 2023). This fragmentation underscores a critical gap in the literature: the lack of integrated forecasting models that holistically combine macroeconomic indicators, technical variables, and sentiment signals derived from financial news. While some recent contributions (e.g., Bâra et al. 2024; Bâra and Oprea 2024) have attempted to merge technical indicators with sentiment analysis, these efforts often depend on generic sentiment extraction tools or overlook the dynamic processing capabilities required for real-time financial contexts. More importantly, very few studies have operationalized sentiment vectors—categorized as positive, negative, or neutral—alongside structured financial and macroeconomic variables within a unified predictive pipeline.

This study aims to fill these critical research gaps by introducing a novel hybrid forecasting framework that integrates FinBERT-derived sentiment analysis features with a comprehensive set of financial, macroeconomic, and temporal indicators. The contribution of this research is twofold.

First, the study employs FinBERT, a domain-specific NLP model tailored for financial text, to extract fine-grained sentiment scores and vectorized features from Bitcoin-related news headlines. These features—comprising daily counts and intensity scores for positive, negative, and neutral news—are systematically merged with structured quantitative data, including Bitcoin's historical price and lagged values (Lag1, Lag2, Lag3), market-based variables (trading volume), financial and macroeconomic indicators, and temporal attributes (year, month, weekday, holiday indicators, week of year, quarter).

Second, these integrated features are utilized in both DL (LSTM, CNN, GRU) and ML (XGBoost, Random Forest, LightGBM, SVR) models to comparatively evaluate their forecasting

performance. By assessing the models before and after the incorporation of FinBERT sentiment features, the study quantifies the added value of sentiment data and identifies the most effective predictive architecture.

An additional innovation lies in the scenario-based “what-if” analysis applied to the best-performing hybrid model, FinBERT-LSTM. This approach enables the simulation of different sentiment and macroeconomic conditions to assess their impact on Bitcoin price forecasts. Finally, the study extends its contribution to volatility modeling by employing GARCH analysis to quantify the asymmetric influence of news sentiment on Bitcoin price volatility, thereby offering insights into how positive, negative, and neutral news flows affect market stability. In sum, this research not only bridges the methodological divide between sentiment analysis and financial modeling but also offers a scalable and replicable forecasting framework. The integration of financial text-specific sentiment vectors, real-time macroeconomic signals, and technical market indicators provides a comprehensive and novel contribution to the Bitcoin forecasting literature. By addressing these critical research gaps, the study offers both theoretical and empirical advancements that are of substantial relevance to academics, investors, and policymakers seeking to understand and navigate the volatile dynamics of cryptocurrency markets.

3 | Methodology

This section introduces the data sets and specifications used for the empirical analysis of the study, and then systematically presents the applied analysis methods and the models used.

3.1 | General Framework of the Proposed Approach

This study proposes an integrated forecasting framework designed to enhance the accuracy of Bitcoin price prediction by combining financial, macroeconomic, temporal, and sentiment-based variables through a hybrid machine learning and deep learning approach. The overall methodology consists of five core stages: data collection and preprocessing, hybrid feature construction, model training and optimization, performance evaluation, and volatility-scenario analysis. To provide a clear overview before the detailed explanations, the main algorithmic flow of the proposed methodology can be summarized as follows:

- Collect and preprocess financial, macroeconomic, temporal, and sentiment-based data.
- Extract sentiment features using FinBERT and reduce their dimensionality through Autoencoder and PCA.
- Construct a hybrid dataset integrating numerical and sentiment variables.
- Apply chronological splitting and rolling-origin cross-validation to ensure time-aware model evaluation.
- Train machine learning and deep learning models with hyperparameter optimization.

- Evaluate performance using multiple regression metrics, followed by volatility and scenario analyses.

At the outset, Bitcoin price data is enriched with lagged values and a diverse set of financial and macroeconomic indicators, including crude oil prices, the US Dollar Index (DXY), gold prices, major stock indices (Dow Jones, S&P 500, Nasdaq), the US 10-year bond yield (TNX), and the volatility index (VIX). In addition, a comprehensive set of calendar-based temporal variables—such as year, month, day, weekday, weekend indicator, quarter, week number, and public holiday markers—is integrated to account for seasonality and structural patterns.

Crucially, to capture the effect of market sentiment, financial news texts collected from multiple platforms including GDELT and Google News are processed through FinBERT, a domain-specific transformer model, to extract positive, negative, and neutral sentiment scores. The 768-dimensional FinBERT embedding vectors derived from each news headline are further refined through dimensionality reduction techniques. A dual-stage method, involving Autoencoder and principal component analysis (PCA), is employed to identify and retain the 10 most salient components that explain the highest variance in sentiment-related semantics. These reduced embeddings are appended to the main dataset, thereby transforming all models into a sentiment-aware hybrid structure.

Hyperparameter tuning is conducted using model-specific strategies: RandomizedSearchCV is employed for ML models such as XGBoost, LightGBM, Random Forest, and SVR, while the Optuna framework is used to optimize DL architectures including LSTM, GRU, and CNN. All datasets are chronologically partitioned into 80% training and 20% testing subsets to preserve temporal integrity, and Min-Max normalization is

applied to ensure feature comparability. To ensure methodological rigor and prevent forward-looking bias, all training and testing procedures were conducted under a chronologically ordered framework. Specifically, the full dataset was partitioned into 80% training and 20% test sets, respecting temporal continuity and avoiding random shuffling. Furthermore, a rolling-origin five-fold time series cross-validation strategy (using the TimeSeriesSplit function from scikit-learn) was employed to assess the generalization performance of each model across temporally disjoint test sets. In each fold, the training window was expanded step-by-step (e.g., Fold 1: 2017–2019 train, 2020 test; Fold 2: 2017–2020 train, 2021 test; etc.), ensuring that only past data was used to predict future values. This design not only maintains the directional structure of financial time series but also strengthens out-of-sample robustness, as performance is evaluated exclusively on data segments never exposed during training. This approach aligns with best practices in time series forecasting literature, where temporal leakage and look-ahead bias are major methodological concerns (Bischl et al. 2023; Min et al. 2024). The evaluation of model performance is carried out using five standard regression metrics—root mean squared error (RMSE), mean squared error (MSE), mean absolute error (MAE), mean absolute percentage error (MAPE), and R^2 —along with a five-fold time series cross-validation procedure to test out-of-sample robustness. Lastly, a GARCH-based volatility modeling phase is incorporated to quantify the asymmetric impact of sentiment types on market volatility. This is complemented by a FinBERT-LSTM-based scenario analysis framework to simulate the effect of hypothetical changes in sentiment under varying market conditions. Collectively, this methodology enables the construction of robust forecasting models that surpass the limitations of traditional approaches by accounting for dynamic and exogenous drivers of Bitcoin price fluctuations. The proposed pipeline is summarized schematically in Figure 1,

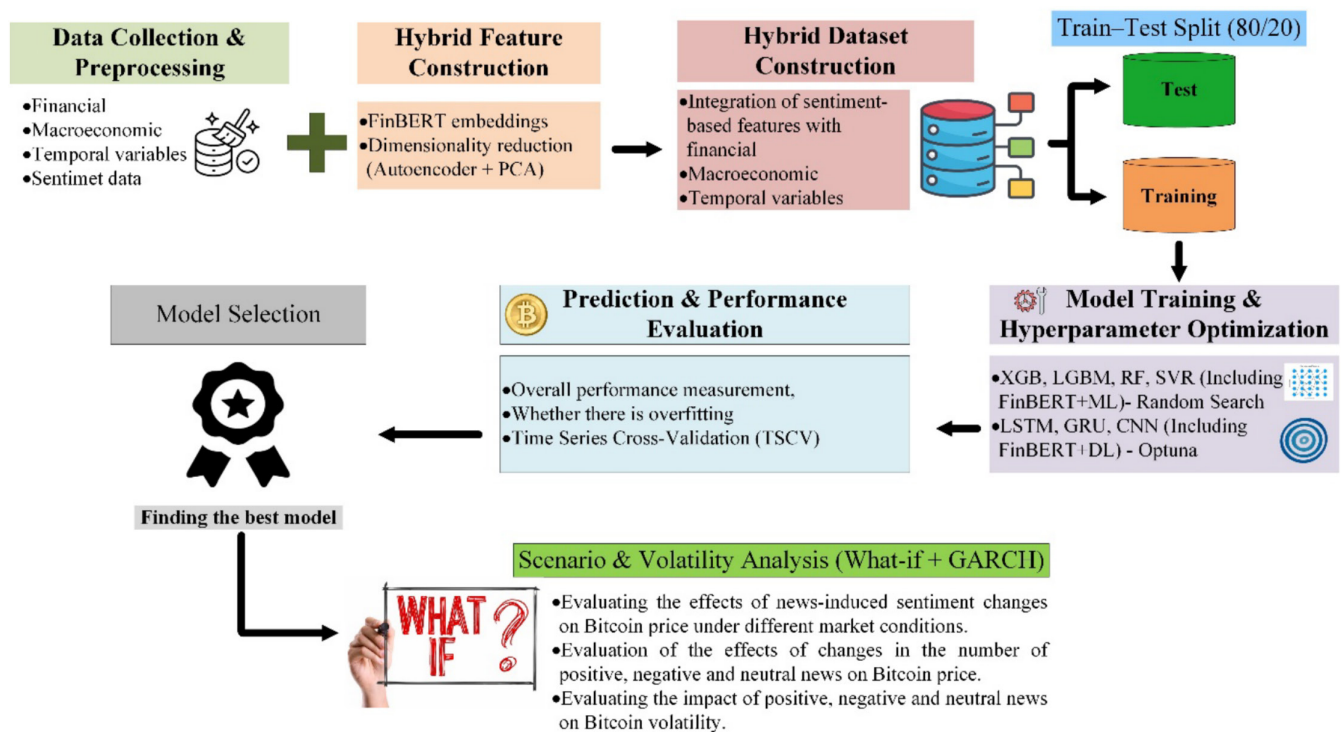


FIGURE 1 | Proposed methodology.

TABLE 3 | Inventory of the independent variables analyzed in the data set and the dependent variable that is the focus of the research.

Kind	Variable	Variable name	Variable description
Financial	x_1	Bitcoin Lag1	Bitcoin closing price with a 1-day lag to capture short-term trends
Financial	x_2	Bitcoin Lag2	Bitcoin closing price with a 2-day lag to capture intermediate short-term trends
Financial	x_3	Bitcoin Lag3	Bitcoin closing price with a 3-day lag to capture extended short-term trends
Macroeconomic	x_4	Oil price	Global crude oil price reflecting macroeconomic conditions
Macroeconomic	x_5	DXY	US dollar index (indicating strength of USD)
Financial	x_6	Gold price	Gold price as a safe-haven asset and economic indicator
Financial	x_7	DJI	Dow Jones Industrial Average, indicator of economic and market conditions
Financial	x_8	S&P 500	S&P 500 Index, capturing market performance of large US firms
Financial	x_9	NASDAQ (IXIC)	NASDAQ Composite Index, heavily weighted towards technology sector
Financial	x_{10}	10-year treasury yield (TNX)	Yield of the US 10-year Treasury bond, reflecting economic outlook and investor risk appetite
Financial	x_{11}	VIX	Volatility index indicating market volatility and investor sentiment
Temporal	x_{12}	Year	Year of observation
Temporal	x_{13}	Month	Month of observation, capturing seasonal patterns
Temporal	x_{14}	Day	Day of observation, capturing daily fluctuations
Temporal	x_{15}	Day of the week	Day of the week, capturing weekly cyclical effects
Temporal	x_{16}	Weekend indicator	Binary indicator (1 = weekend, 0 = weekday)
Temporal	x_{17}	Quarter of the year	Quarter of the year (Q1–Q4) to capture quarterly effects
Temporal	x_{18}	Week of the year	Week number of the year, useful for weekly seasonal variations
Temporal	x_{19}	Holiday indicator	Indicator for public holidays (yes = 1, no = 0), capturing their impact on trading behavior
Financial	y	Bitcoin price (dependent variable)	Represents daily Bitcoin price in dollars.

which outlines the sequential flow of operations and their integration points across model development.

3.2 | Dataset and Preprocessing

In this study, a two-stage modeling strategy is adopted for forecasting Bitcoin price movements. The first stage involves the construction of a dataset that integrates financial, macroeconomic, and temporal variables, covering the period from November 2017 to January 2025. As illustrated in Table 3, the financial indicators include three lagged values of Bitcoin closing price (x_1 – x_3), which are used to capture short-term and medium-term autoregressive effects. The financial and macroeconomic variables (x_4 – x_{11}) include daily observations of crude oil price, the DXY, gold price, Dow Jones Industrial Average (DJI), S&P 500, NASDAQ (IXIC), the TNX, and the

VIX, all of which reflect market-level macroeconomic dynamics. In addition, a set of calendar-based temporal variables (x_{12} – x_{19}) such as year, month, day of the week, quarter, and holiday indicators are included to account for periodic and seasonal fluctuations.

In the second stage, the dataset is extended to include news-based sentiment variables extracted from Bitcoin-related headlines collected daily from financial news sources such as Google News and GDELT, covering the same time period (November 2017–January 2025). The sentiment analysis is conducted using the FinBERT model, a financial domain-specific transformer-based NLP model, which generates sentiment classifications (positive, negative, neutral), overall sentiment scores, and high-dimensional embedding vectors representing the semantic structure of the headlines. The sentiment variables integrated into the dataset are presented in Table 4.

TABLE 4 | News-based sentiment analysis variables included in the dataset using the FinBERT model.

Kind	Variable	Variable name	Variable description
Number of daily news	x_{20}	Positive news	Daily number of positive news published on Bitcoin
Number of daily news	x_{21}	Negative news	Daily number of negative news published on Bitcoin
Number of daily news	x_{22}	Neutral news	Daily number of neutral news published on Bitcoin
Total number of daily news	x_{23}	News count	Total number of daily news published on Bitcoin
News sentiment data	x_{24}	FinBERT sentiment	Sentiment score obtained from the news of the relevant day
FinBERT embedding vectors	$x_{25} - x_{34}$	FinBERTDim (vector numbers)	Embedding vectors extracted from news headlines with FinBERT model

In order to avoid multicollinearity and reduce redundancy among the 768-dimensional FinBERT embedding vectors, a two-step dimensionality reduction method combining Autoencoder and PCA is employed. The top 10 components with the highest explained variance (92.64%) are retained and incorporated into the final dataset. This approach ensures both feature efficiency and semantic representation consistency across news content. In addition, the detailed rationale and implementation steps for selecting these top 10 features using Autoencoder and PCA are comprehensively explained in Section 3.3.

For model training and testing, both the baseline and the hybrid datasets are split chronologically using an 80/20 ratio to preserve temporal integrity and prevent data leakage. No random shuffling is performed in order to maintain the time series structure of the Bitcoin price data. All input variables are normalized using Min-Max scaling, which maps the values into the [0, 1] interval, enhancing convergence during training and reducing the dominance of high-magnitude features. This technique is preferred for its simplicity, efficiency, and effectiveness in time series forecasting problems (Chimphlee and Chimphlee 2023; Elshewey et al. 2023). With the integration of news-based sentiment information into the feature space, the resulting hybrid dataset enables a comparative evaluation of traditional macro-financial models and those enriched with NLP-driven sentiment dimensions. The Min-Max scaling formula is shown in Equation (1):

$$x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (1)$$

3.3 | Feature Engineering and Hybrid Construction

To improve the predictive capacity of the proposed hybrid framework, this study combines traditional financial, macroeconomic, and temporal variables with sentiment-based features extracted from news texts. While the first group forms the classical foundation of time series modeling, the second group reveals the textual and emotional dimensions of financial communication, providing an enriched feature space for forecasting. For this purpose, Bitcoin-related news headlines covering the period from November 2017

to January 2025 were collected from various financial news sources such as GDELT, Google News, Reuters, Bloomberg, and CoinDesk. These headlines were cleaned using NLP techniques such as lowercase conversion, removal of multilingual stop words, expansion of domain-specific stop words, and tokenization to ensure semantic consistency across sources and languages.

Sentiment analysis was performed using the FinBERT model, a transformer-based language model pre-trained on financial texts. Three sentiment scores were calculated for each news headline: positive, negative, and neutral. To make these scores functional in a time series prediction format, daily totals were calculated, including the total number of news articles and the number of articles per sentiment class. Additionally, daily positive and negative sentiment scores were compared to derive a categorical indicator called “FinBERT Sentiment.” This indicator preserves the emotional direction while reducing comment complexity, yielding values of 1 (positive), 0 (neutral), or −1 (negative).

In addition to separate sentiment scores, continuous FinBERT embedding vectors were obtained for each headline. Each sentence was encoded into a 768-dimensional vector capturing semantic information and contextual nuances. The daily average of these vectors was taken to create daily-level representations. Using high-dimensional vectors directly in prediction models increases computational costs and may lead to overfitting (Jadia 2023; Suganya et al. 2024). To address this issue, a two-stage dimension reduction strategy was applied. The general dimension reduction process is summarized schematically in Figure 2. This figure visualizes how Autoencoder compression and PCA-based selection work together to isolate the most informative FinBERT features. The relationship between autoencoders and PCA is noteworthy for their complementary roles in dimension reduction tasks. While PCA uses a linear transformation approach to reduce dimensions based on maximizing variance, autoencoders can model non-linear relationships in data through layered architectures and non-linear activation functions (Ibebuchi et al. 2024). Additionally, recent studies have shown that integrating PCA techniques into autoencoder architectures can enhance feature extraction capabilities and enable the extraction of more refined compressed representations from input data (Luchi et al. 2023). This integration highlights the

versatility of autoencoders in achieving lower dimensions while preserving the important features of the original data.

First, an Autoencoder architecture consisting of five encoding layers ($768 \rightarrow 128 \rightarrow 64 \rightarrow 64 \rightarrow 32 \rightarrow 16$) and symmetric decoding layers was trained for 50 epochs using the Adam optimizer and MSE as the objective function to minimize reconstruction loss. This unsupervised compression provided a compact 16-dimensional representation that preserved the underlying latent features. The loss curve presented in Figure 3 demonstrates the model's convergence and the steady decrease in reconstruction error over time.

The decline in the reconstruction error over training epochs, as illustrated in Figure 3, confirms that the Autoencoder effectively captured the latent semantic structure of FinBERT embeddings. To perform a more detailed analysis of the 16-dimensional representation generated by the Autoencoder, PCA was applied, and the 10 components with the highest explained variance

were extracted. This confirms that the Autoencoder effectively learned to compress and reconstruct the FinBERT embedding vectors with minimal information loss. Monitoring reconstruction loss is crucial to confirm that the compressed representations preserve the semantic structure of the original news data and are therefore suitable for downstream prediction tasks. This approach balances the Autoencoder's nonlinear representation capability with PCA's interpretability and orthogonality. The scree plot showing the cumulative explainable variance of the PCA output is provided in Figure 4 to validate the selection of the top 10 components. These 10 features were added to the main dataset to form a final hybrid feature set combining numerical, temporal, and emotional dimensions.

As illustrated in Figure 4, the scree plot demonstrates the cumulative explained variance of the FinBERT embedding vectors after applying PCA to the compressed 16-dimensional representations obtained via Autoencoder. The first 10 principal components collectively account for 92.64% of the total variance, indicating a

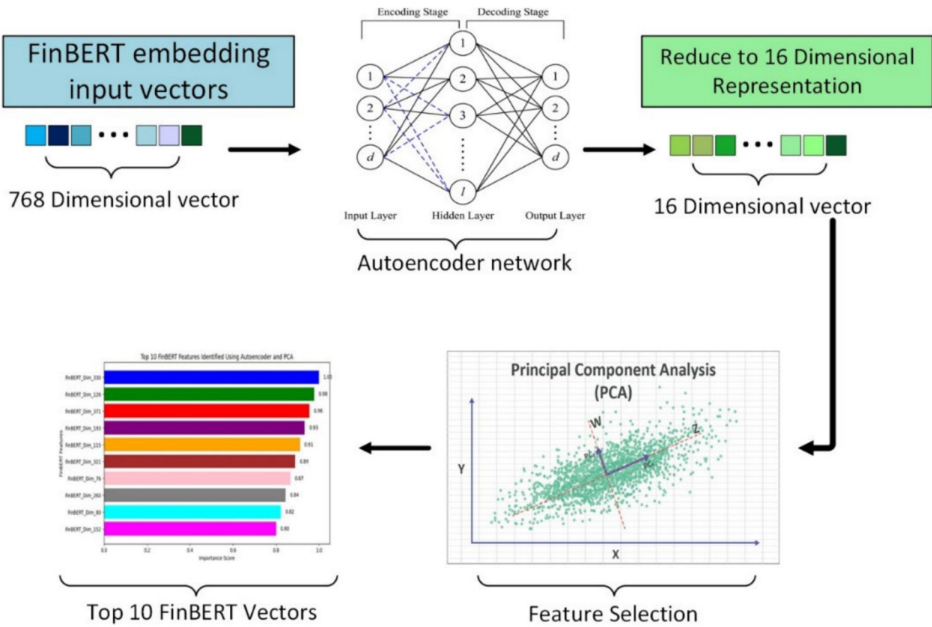


FIGURE 2 | Schematic overview of the dimension reduction pipeline integrating Autoencoder compression and PCA-based feature selection.

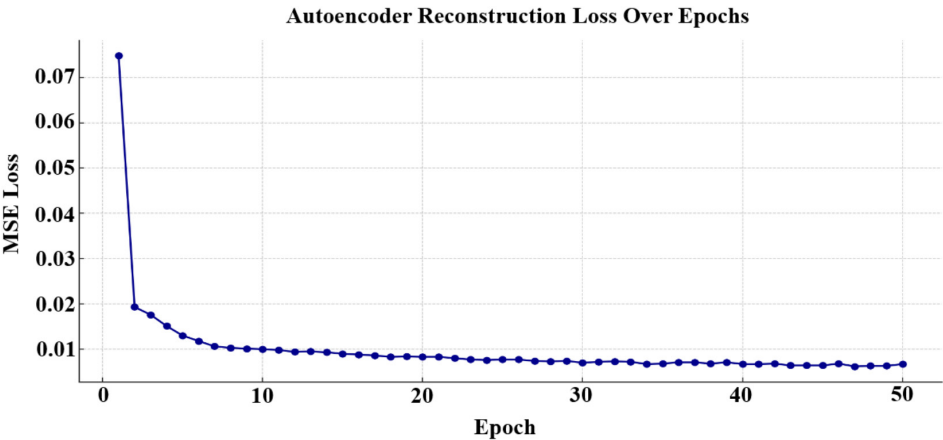


FIGURE 3 | Autoencoder training curve: MSE-based reconstruction loss over 50 epochs.

highly effective dimensionality reduction. The steep ascent of the cumulative variance curve in the initial components and the clear elbow point further validate the selection of the top 10 components. The gray vertical dashed line in the figure highlights the cutoff point at the 10th component, beyond which the marginal gain in explained variance diminishes substantially. This result confirms that the most informative and non-redundant semantic features have been retained, ensuring that essential structural information from the original high-dimensional FinBERT embeddings is preserved for downstream prediction tasks. These 10 features were added to the main dataset to form a final hybrid feature set that combines numerical, temporal, and emotional dimensions. Figure 5 provides an interpretable foundation by showing the selected top 10 features and their importance scores for inclusion in the final hybrid dataset. The integrated dataset structure in the proposed methodology enables the comparative evaluation of ML and DL models in both traditional and hybrid configurations. The

final configuration process and a summary of the hybrid dataset composition are presented in Section 3.7.

As shown in Figure 5, among the 768 original FinBERT dimensions, dimensions such as “Dim_330” and “Dim_126” consistently yield the highest information scores, indicating their critical role in encoding sentiment-relevant features.

3.4 | Model Architecture and Hyperparameter Optimization

This section presents the model implementation strategy and optimization framework adopted to evaluate the forecasting performance of the proposed hybrid dataset. Here we focus on the methodological contributions of the study, in particular, the structured integration of time-sensitive training and validation

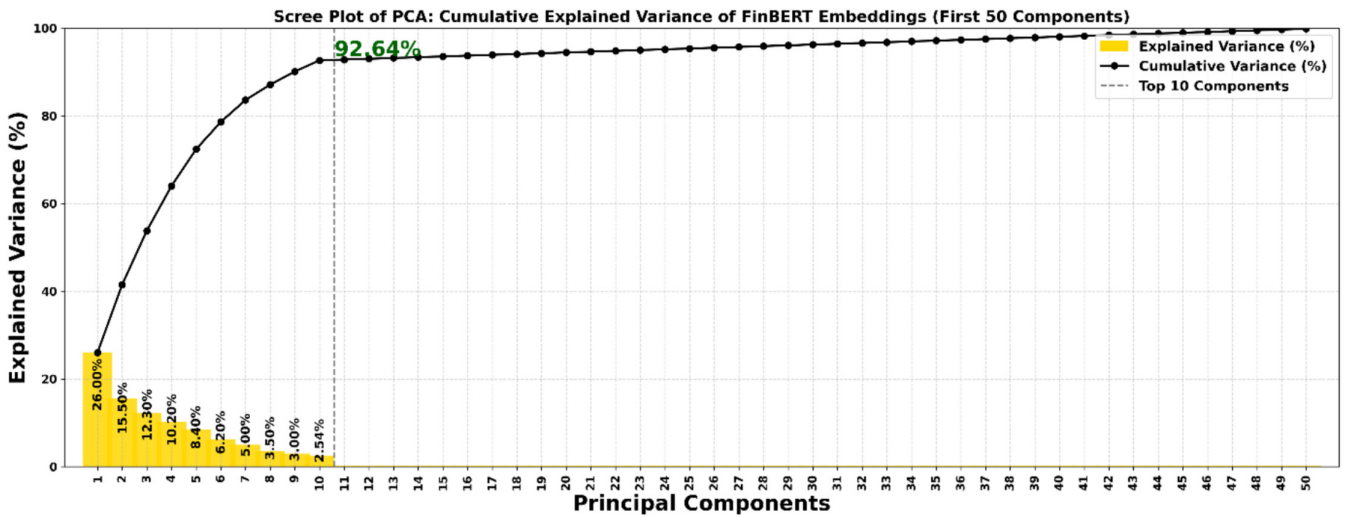


FIGURE 4 | Scree plot of PCA output.

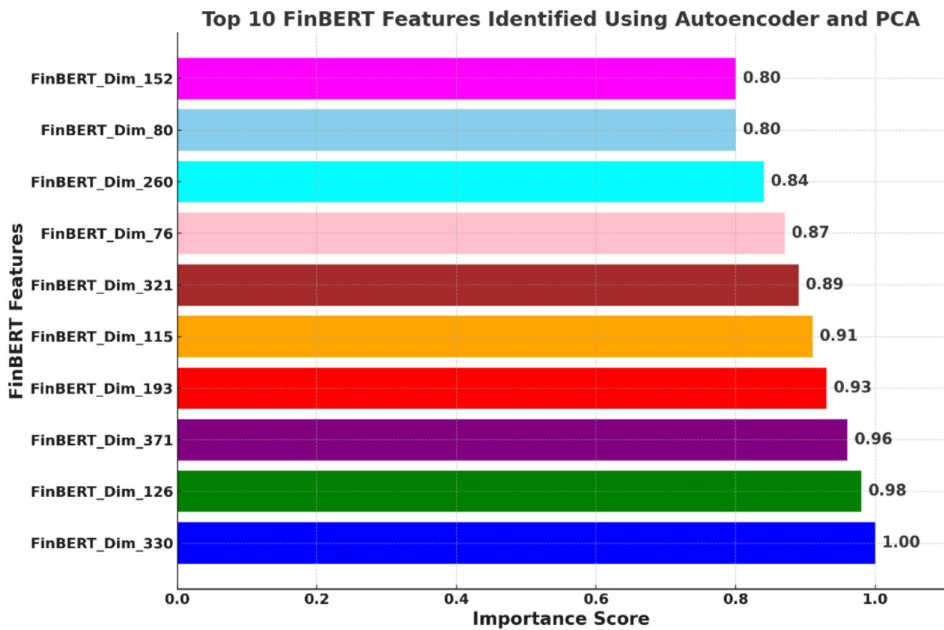


FIGURE 5 | Top 10 FinBERT vectors features importance.

procedures. In particular, the chronological data partitioning process, the structuring of time series-specific cross-validation folds and the measures used to ensure out-of-sample robustness are highlighted. This methodological formalization strengthens the empirical reliability of the results and aligns the modeling protocol with the temporal nature of the forecasting task.

3.4.1 | Machine Learning and Deep Learning Methods

This study applies various ML and DL models to evaluate the predictive capacity of a proposed hybrid feature set for predicting Bitcoin price movements. The study emphasizes the methodological integration of different modeling paradigms, the evaluation strategy, and the performance comparison. To ensure realistic predictions and prevent forward bias, all models were trained and evaluated using a time series-aware cross-validation strategy. A rolling-root (expanding window) 5-fold cross-validation approach was adopted, where the training sets were chronologically expanded and tested on subsequent unseen data. This design accounts for the temporal structure of the data and ensures that future values do not leak into the training set, providing more reliable out-of-sample performance evaluations.

During the modeling process, ML algorithms including XGB, Random Forest, LightGBM, and SVR were used. These models were selected for their ability to determine the importance of features in time series predictions, model non-linear relationships, and control overfitting. In particular, gradient boosting-based models such as XGB and LightGBM have been shown to perform well on high-dimensional datasets and complex feature relationships in various studies (Li et al. 2021; Gurcan and Soylu 2024; Mizrahi et al. 2024; Xing et al. 2024) and are expected to be effective in predicting the Bitcoin price in this context. Additionally, RNNs such as LSTM and GRU are used in time series analysis due to their ability to learn long-term dependencies. Both models can preserve long-term dependencies and have the ability to learn from past data through memory mechanisms (Wang et al. 2023; Ren et al. 2023; Carrera and Kim 2024). Therefore, they are preferred for modeling the historical trends of the Bitcoin price. The CNN model is used to identify short-term patterns in price series and discover relationships between news sentiment scores and price changes. The unique architecture of CNNs, designed to capture spatial hierarchies and local dependencies between data, makes them particularly suitable for capturing the dynamic trends typically found in financial time series data (Wang 2024; Hoa et al. 2025). The nonlinear relationships required in financial markets can be effectively modeled using CNNs, as they can automatically learn important representations from input data without requiring extensive feature engineering (Zheng 2023; Zhao et al. 2024). These models were selected not for their novelty but for their proven success in previous financial forecasting literature and their distinct strengths in capturing nonlinear structures, interactions, and sequential dependencies.

Importantly, the selection of these models is also directly aligned with the empirical properties of our hybrid dataset. The Bitcoin price data exhibit strong autocorrelation and temporal dependencies, which justify the use of recurrent architectures such as LSTM and GRU. At the same time, sudden sentiment-driven shocks and local fluctuations are frequent in the series, making

CNN particularly suitable for capturing short-term structural changes. The integration of FinBERT embeddings with macro-financial and temporal indicators produces a high-dimensional and nonlinear feature space, for which tree-based ensemble methods (XGB, LightGBM, RF) are well suited given their capacity to model complex interactions and provide interpretable feature importance measures. Finally, given the pronounced volatility of cryptocurrency markets, SVR offers robustness through its margin-based learning, ensuring stable performance under conditions of high variance. Thus, each model aligns with a specific property of the dataset, ensuring that the comparative evaluation is not only methodologically diverse but also empirically well grounded.

In summary:

- Tree-based ensemble methods were selected for their ability to process high-dimensional inputs and provide feature importance metrics.
- SVR was used because of its margin-based error tolerance, which generally provides an advantage in volatile market conditions.
- RNN-based methods are effective in modeling long-term temporal dependencies, which are crucial for sequential data such as the Bitcoin price.
- CNN has been integrated to detect local patterns and structural changes in data, especially when enhanced with FinBERT-based sentiment features.

In the study, the Bitcoin price was re-estimated by ML and DL models using a new hybrid dataset created by combining the variables in the original dataset with variables such as embedding vectors obtained from the FinBERT model, daily sentiment score, daily positive, negative, neutral and total number of news. As a result of this integration, FinBERT-LSTM, FinBERT-CNN, FinBERT-GRU, FinBERT-XGB, FinBERT-RF, FinBERT-LightGBM and FinBERT-SVR hybrid models were developed and the impact of sentiment analysis variables on forecasting performance was compared. The general structure of the hybrid models is formulated as shown in Equation (2).

$$y_t = f(X_t, S_t, E_t, Z_t) \quad (2)$$

where X_t denotes financial, macroeconomic and temporal variables, S_t denotes daily news sentiment scores (Positive, Negative, Neutral-FinBERT Sentiment), E_t denotes FinBERT embedding vectors, and Z_t denotes the number of positive, negative, and neutral news stories. The purpose of these models is to enhance the impact of sentiment analysis on the Bitcoin price and to quantify how news-based variables affect forecast accuracy.

3.4.2 | Hyperparameter Optimization

Hyperparameter optimization is a critical step for achieving high predictive accuracy in Bitcoin price forecasting models. In this study, two complementary optimization strategies were employed in accordance with model type and complexity:

randomized search cross-validation for classical ML algorithms and Optuna-based optimization for DL architectures. Importantly, all optimization procedures were conducted using five-fold time series cross-validation, which respects the temporal order of data and avoids look-ahead bias. This approach ensures that performance metrics reflect realistic out-of-sample robustness by testing each fold exclusively on future (unseen) data segments, as recommended in recent methodological literature (Vardhan et al. 2023; Bischl et al. 2023).

To determine the optimal combination of hyperparameters for each ML model, the RandomizedSearchCV method was utilized with 100 distinct parameter combinations. Unlike Grid Search, which exhaustively evaluates all combinations in a predefined space, randomized search samples random values from predefined distributions, making it computationally more efficient while still capable of identifying near-optimal solutions (Japa et al. 2023; Vardhan et al. 2023). The models were evaluated using the negative mean squared error as the scoring metric. As shown in Table 5, the best-performing hyperparameters were selected based on their validation error across time-aware CV folds.

For DL models such as LSTM, GRU, and CNN, hyperparameter tuning was performed using the Optuna framework, which offers a dynamic and efficient approach to high-dimensional optimization. Optuna's *define-by-run* paradigm enables flexible search space definition during runtime, in contrast to static grid-based methods (Prasanna et al. 2023). Moreover, Optuna employs the tree-structured Parzen estimator (TPE) as its core optimization algorithm, which models the probability distribution of good-performing trials based on historical performance and actively guides the search process (Gabriel and Anbarasi 2023). Another key feature leveraged in this study is Optuna's pruning mechanism, which monitors validation loss during training and terminates unpromising trials early. This significantly reduces computational cost—especially important for DL models with complex architectures and long training times (Klaar et al. 2023). For each DL model, 100 independent trials were conducted within predefined

hyperparameter ranges. Each trial was evaluated using time-aware validation, ensuring that the tuning process respects the chronological order of Bitcoin data and enhances out-of-sample generalization (Monshi et al. 2021; Min et al. 2024). The final architecture for LSTM and GRU consisted of three sequential layers, whereas the CNN model utilized four convolutional blocks. Optuna optimized parameters such as the number of units or filters per layer, dropout rates, learning rate, batch size, number of epochs, and optimizer type. The results are summarized in Table 6.

3.5 | Software and Hardware

In this study, data processing, model training and performance evaluation processes were performed on a computer with a high-end AMD Ryzen processor, a Sapphire Nitro+ Radeon RX 7900 XTX graphics card with 32GB GPU memory and 32GB DDR5 RAM under the Windows 11 operating system. All analyses were conducted using the Jupyter Notebook, which offers a flexible and interactive working environment. The Python programming language was preferred for training ML and DL models, and a wide library ecosystem was integrated into the data processing, model building, visualization, and evaluation stages. Pandas, NumPy, and Scikit-learn libraries were utilized for data preprocessing, and Statsmodels was used for time series data analysis. In NLP-based sentiment analysis processes, tools such as Transformers (Hugging Face), FinBERT, NLTK, and WordCloud were used to analyze features derived from news data. Scikit-learn, XGBoost, and LightGBM libraries were used for the implementation and optimization of ML models, and hyperparameter optimization was carried out with RandomizedSearchCV. DL models were created using TensorFlow and Keras libraries and hyperparameter optimization was performed with the Optuna algorithm. Graphical summaries were created using Matplotlib and Seaborn for all visualization and presentation of analysis results. Thanks to the high processing power provided by the software and hardware infrastructure and optimized library integrations, data analysis and model training processes were carried out quickly

TABLE 5 | Hyperparameter optimization results for ML models.

Model	Hyperparameter	Search range	Best parameters
XGB	N estimators, learning rate, subsample, max depth, colsample bytree, min child weight, gamma	50–500, 0.01–0.6, 0.3–0.7, 1–10, 0.5–0.9, 1–10, 0–0.5	N estimators: 308, learning rate: 0.249, subsample: 0.664, max depth: 6, colsample bytree: 0.742, min child weight: 6, gamma: 0.011
RF	N estimators, max depth, min samples split, min samples leaf, max features, bootstrap	50–500, 5–20, 2–10, 1–5, auto/sqrt/log2, true/false	N estimators: 360, max depth: 8, min samples split: 4, min samples leaf: 3, max features: sqrt, bootstrap: true
SVR	C, epsilon, gamma, kernel	0.1–100, 0.01–0.5, scale/auto, linear/rbf/poly/sigmoid	C: 20.652, epsilon: 0.027, gamma: auto, kernel: linear
LGBM	N estimators, learning rate, num leaves, max depth, min child samples, subsample, colsample bytree	50–500, 0.01–0.3, 10–100, 3–15, 5–50, 0.5–1, 0.5–1	N estimators: 207, learning rate: 0.050, num leaves: 28, max depth: 4, min child samples: 46, subsample: 0.898, colsample bytree: 0.569

TABLE 6 | Hyperparameter optimization results for DL models.

Model	Hyperparameter	Search range	Best parameters
LSTM	LSTM1 units, LSTM2 units, LSTM3 units, dropout rate, learning rate, batch size, epochs, optimizer	50–300, 50–200, 25–100, 0.1–0.5, 0.00001–0.01, 16–64, 10–100, Adam/RMSprop/SGD	LSTM1: 200, LSTM2: 100, LSTM3: 50, dropout rate: 0.3, learning rate: 0.0001, batch size: 64, epochs: 50, optimizer: Adam
GRU	GRU1 units, GRU2 units, GRU3 units, dropout rate, learning rate, batch size, epochs, optimizer	50–300, 50–200, 25–100, 0.1–0.5, 0.00001–0.01, 16–64, 10–100, Adam/RMSprop/SGD	GRU1: 200, GRU2: 100, GRU3: 50, dropout rate: 0.3, learning rate: 0.0001, batch size: 64, epochs: 50, optimizer: Adam
CNN	CNN1 filters, CNN2 filters, CNN3 filters, CNN4 filters, kernel size, stride, dropout rate, learning rate, batch size, epochs, optimizer	32–512, 32–256, 32–128, 32–64, 3–7, 1–3, 0.1–0.5, 0.00001–0.01, 16–64, 10–100, Adam/RMSprop/SGD	CNN1: 512, CNN2: 256, CNN3: 128, CNN4: 64, kernel size: 3, stride: 1, dropout rate: 0.3, learning rate: 0.0001, batch size: 64, epochs: 50, optimizer: Adam

and efficiently, and successful prediction performances were achieved on large-scale data sets.

3.6 | Model Performance Evaluation Metrics

In this study, five different performance metrics such as RMSE, MSE, MAE, MAPE, and R^2 Score are used to evaluate the accuracy and reliability of Bitcoin price prediction models. These metrics were chosen to measure the agreement of the predicted Bitcoin price with actual prices and to compare the overall performance of the models. The mathematical formulations of these metrics are shown in Equations (3), (4), (5), (6) and (7), respectively.

RMSE is calculated as the square root of the mean square of the squares of the differences between the predicted values and the actual values. This metric is important in highlighting the magnitude of forecast errors and measures the model's error sensitivity, and the smaller the RMSE value, the better the model's forecast performance (Chicco et al. 2021).

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (3)$$

MSE is the mean of the squares of the differences between the predicted values and the actual values. It gives more weight to larger errors, so it is useful for detecting large deviations. Similar to RMSE, a low MSE indicates the success of the model (Kaba and Suzer 2021).

$$MSE = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (4)$$

MAE is calculated by averaging the absolute values of the differences between predicted values and actual values. This metric directly measures the magnitude of prediction errors and is a more balanced measure as it does not square the deviations. A lower MAE value indicates that the model produces more accurate predictions (Zhou et al. 2023).

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (5)$$

MAPE is a metric that gives the average of prediction errors in percent. This metric is useful for understanding the error rate and comparing models at different scales. A low MAPE value indicates that the model's predictions are closer to the true values (Tang et al. 2023).

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{y_i} \right| \quad (6)$$

R^2 is a statistic that shows how much of the variance in the dependent variable is explained by the model. By measuring the explanatory power of the model, it shows how well the predicted prices match the actual prices (Dönmez and Aslan 2023).

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (7)$$

where $\bar{y}$ is the average of the actual values. The R^2 score approaching 1 indicates that the predictive power of the model increases.

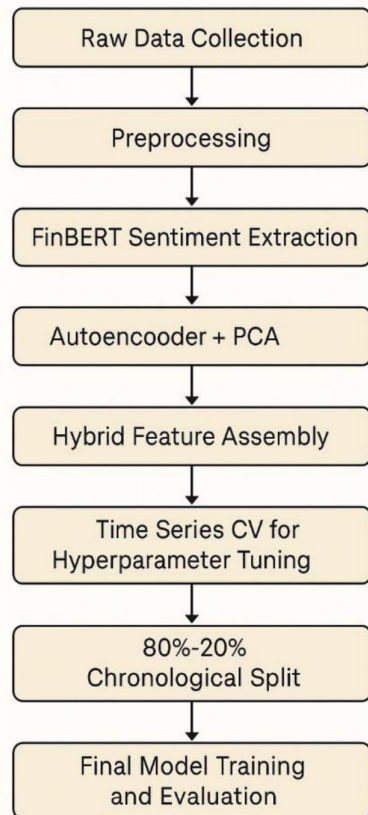
3.7 | Final Hybrid Dataset Construction and Implementation Flow

As detailed in Section 3.1, the proposed methodology integrates multiple layers of financial, macroeconomic, temporal, and sentiment-based variables to form a hybrid dataset tailored for Bitcoin price forecasting. This section provides a concise yet systematic overview of the final feature space and implementation flow adopted prior to model training.

To ensure a structured and semantically rich input configuration, four distinct feature groups were constructed: (i) financial, macroeconomic, and temporal variables (denoted as X_f), (ii)

TABLE 7 | Structural components of the final hybrid feature set.

Feature group	Description	Variables
X_t	Financial, macroeconomic, and temporal indicators	Crude oil price, USD Index, S&P500, DJIA, Nasdaq, VIX, US10Y yield, lagged prices, RSI, MACD, year, month, weekday, holiday, etc.
S_t	Sentiment scores from FinBERT	Daily FinBERT-based positive, negative, neutral scores
E_t	Reduced-dimensional FinBERT embeddings	The 10 best features selected from 768-dimensional vectors using the Autoencoder + PCA method
Z_t	News volume statistics	Daily count of positive, negative, neutral news articles

**FIGURE 6** | Implementation flow of the hybrid forecasting framework.

FinBERT-derived sentiment scores (S_t), (iii) dimensionality-reduced embedding vectors (E_t), and (iv) the frequency of sentiment-specific news articles (Z_t). These feature blocks were jointly integrated to serve as the model input. Table 7 summarizes the components and characteristics of each feature group.

All input variables were scaled using Min-Max normalization, and the final hybrid dataset was structured chronologically to preserve the temporal order of observations and prevent forward-looking bias. Hyperparameter optimization was conducted using five-fold time series cross-validation exclusively on the training set, ensuring that model tuning respected the temporal structure of the data. In this process, each fold's validation segment was strictly posterior to its training segment, thereby simulating realistic out-of-sample

conditions. After optimal hyperparameters were identified, the full dataset was partitioned chronologically into 80% training and 20% test sets. Final model training was performed on the entire training set using the best hyperparameter configuration derived from time series cross-validation. Performance evaluation was then conducted on the temporally disjoint test set, which had never been exposed to the model during training or hyperparameter tuning. This protocol guarantees out-of-sample validity and minimizes temporal leakage. The complete implementation workflow, including data integration, feature transformation, time-aware validation, and performance assessment, is schematically illustrated in Figure 6.

This modular structure ensures that both domain-driven and sentiment-derived predictors are jointly optimized to enhance the forecasting accuracy of Bitcoin price models, while maintaining strict temporal consistency and methodological rigor.

4 | Findings

In this section, we comprehensively evaluate the effectiveness of the proposed methodology on Bitcoin price prediction. First, we compare the forecasting performance of models using financial, macroeconomic and temporal indicators and hybrid models integrating sentiment analysis variables. By training ML and DL models with both datasets, we observe that the models with integrated sentiment analysis features improve the prediction accuracy. Then, the results of the model with the highest prediction performance are detailed. Using this model, what-if scenario analyses are performed and the effects of news-induced sentiment changes on the Bitcoin price under different market conditions are evaluated. In addition, in order to analyze the relationship between news sentiment and Bitcoin price volatility, the GARCH model is used to examine the effects of positive, negative and neutral news on volatility.

4.1 | Performance Comparison of Models

The performance of the forecasting models used in this study is evaluated in two stages: before and after the integration of sentiment analysis variables. In the first stage, the forecasting accuracy of the models trained with the traditional dataset containing only financial, macroeconomic and temporal variables is analyzed. In the second stage, the performance of the hybrid models trained with the extended dataset created by integrating

all variables obtained from sentiment analysis is compared. For the first stage, the training, testing and cross-validation performances of each model were evaluated using metrics such as R^2 , MAE, MAPE, MSE and RMSE. The metric results obtained from the training, testing, and cross-validation sets for DL and ML models are shown in Table 8.

When the modeling results based on traditional financial and macroeconomic indicators and temporal variables in Bitcoin price forecasting shown in Table 8 are analyzed, it is observed that DL models generally provide higher prediction accuracy compared to ML models. DL models such as LSTM, CNN, and GRU performed successful predictions with high R^2 values and low error metrics (RMSE, MAE, MAPE) in the test set. In particular, the LSTM model stood out as the most successful model in Bitcoin price prediction with $R^2=0.9548$, RMSE=0.0439 and MAPE=4.66% in the test set. While the CNN model also provided high accuracy with $R^2=0.9393$, RMSE=0.0509, and MAPE=6.68% in the test set, the GRU model showed a lower overall performance compared to other DL models with $R^2=0.8639$ and RMSE=0.0762 in the test set, although it gave successful results in the training set. When the cross-validation results are examined, it is seen that DL models can generalize consistently with their training and test performances. LSTM was the most stable model in the cross-validation set

with $R^2=0.9356$ and RMSE=0.0241, while the CNN model performed similarly with $R^2=0.9340$ and RMSE=0.0237. The GRU model showed a lower generalization performance than the other DL models with $R^2=0.8989$ and RMSE=0.0314 in the cross-validation set. When the results were evaluated in terms of ML models, the prediction accuracy of the models was significantly lower compared to the DL models. The XGB, RF and LGBM models produced unstable results with low test R^2 values, indicating that ML models do not adequately capture data complexity in Bitcoin price prediction. However, the SVR model offered the highest prediction accuracy among both DL and ML models with $R^2=0.9748$, RMSE=0.0329, and MAPE=3.81% in the test set. Furthermore, the time series cross-validation (tcv) results also showed a strong generalization performance with $R^2=0.9116$. These results show that SVR can effectively learn the non-linear relationships in the dataset and make successful predictions without over-parameterization (overfitting). As a result, the SVR model was the most successful model with the lowest error rates and the highest R^2 scores. The LSTM model ranked second, offering the best forecasting performance among the DL models. Therefore, SVR and LSTM models stand out as the most effective methods for Bitcoin price prediction. Although these results show that DL models in general have stronger forecasting capacity compared to ML models, SVR model proves to be particularly outstanding in Bitcoin price prediction. However,

TABLE 8 | Performance metric results for training, testing and cross-validation of all models (without FinBERT sentiment analysis variables).

Model	Set	R^2	MAE	MAPE	MSE	RMSE
LSTM	Training	0.9858	0.0120	0.0678	0.0003	0.0186
	Test	0.9548	0.0305	0.0466	0.0019	0.0439
	Time series CV (5-Fold)	0.9356	0.0174	0.0572	0.0008	0.0241
GRU	Training	0.9861	0.0120	0.0658	0.0003	0.0183
	Test	0.8639	0.0589	0.0889	0.0058	0.0762
	Time series CV (5-Fold)	0.8989	0.0243	0.0673	0.0017	0.0314
CNN	Training	0.9904	0.0099	0.0568	0.0002	0.0152
	Test	0.9393	0.0407	0.0668	0.0026	0.0509
	Time series CV (5-Fold)	0.9340	0.0181	0.0543	0.0008	0.0237
XGB	Training	0.9957	0.0065	0.0458	0.0001	0.0102
	Test	0.5283	0.0758	0.0936	0.0203	0.1424
	Time series CV (5-Fold)	0.8216	0.0235	0.0504	0.0050	0.0392
RF	Training	0.9976	0.0044	0.0221	0.0001	0.0076
	Test	0.2140	0.1197	0.1592	0.0335	0.1831
	Time series CV (5-Fold)	0.7233	0.0325	0.0538	0.0081	0.0464
LGBM	Training	0.9968	0.0055	0.0304	0.0001	0.0087
	Test	0.4399	0.0863	0.1091	0.0239	0.1546
	Time series CV (5-Fold)	0.7951	0.0253	0.0471	0.0058	0.0411
SVR	Training	0.9932	0.0100	0.1017	0.0002	0.0128
	Test	0.9748	0.0247	0.0381	0.0011	0.0329
	Time series CV (5-Fold)	0.9116	0.0132	0.0687	0.0004	0.0169

since all models are trained based on financial, macroeconomic, and temporal indicators only, a critical research question is how the hybrid use of news sentiment analysis features with financial and macroeconomic indicators can affect model prediction accuracy. Therefore, in the next step, sentiment analysis variables obtained with the FinBERT model are integrated into the dataset and analyzed how the forecasting performance of the models changes. Table 9 shows these results.

The findings presented in Table 9 show that the integration of news sentiment significantly improves the forecasting accuracy of all models. In particular, FinBERT-based DL models perform better forecasts with significantly higher R^2 and lower error metrics compared to traditional DL and ML models. When the training set performances of the models are analyzed, it is seen that both DL and ML models offer high accuracy, but some models tend to overfitting. For example, although models such as FinBERT-RF and FinBERT-LGBM had very high R^2 values in the training set, they generalized poorly with low R^2 values in the test set. When the test set performance of the models is analyzed, FinBERT-LSTM model showed the best prediction performance among all models with $R^2=0.9808$, RMSE = 0.0286, and MAPE = 3.73% in the test set. FinBERT-GRU and FinBERT-CNN models

also produced strong results in the test set, but lagged behind FinBERT-LSTM. The FinBERT-SVR model was the most successful among the ML models with $R^2=0.9787$ and RMSE = 0.0303, but lagged behind FinBERT-LSTM when compared to DL models. This model had a successful forecasting capacity even before the integration of news sentiment variables, but its performance improved with the addition of news variables. The FinBERT-XGB model performed poorly in the test set with $R^2=0.4933$, while the FinBERT-RF and FinBERT-LGBM models had low forecasting accuracy despite the addition of news sentiment ($R^2=0.0882$ for FinBERT-RF and $R^2=0.4481$ for FinBERT-LGBM). This suggests that models such as FinBERT-RF and FinBERT-LGBM do not make sufficient use of sentiment analysis variables. Looking at the cross-validation results, the FinBERT-LSTM model provided the highest generalization performance, followed by FinBERT-SVR. This suggests that the integration of news sentiment variables provides stable predictions, especially for DL models. FinBERT-SVR is the most successful ML model with $R^2=0.9308$, while FinBERT-XGB and FinBERT-LGBM models have relatively low accuracy rates. In particular, the FinBERT-RF model also exhibited low generalization capacity in the cross-validation phase and did not benefit sufficiently from news sentiment variables.

TABLE 9 | Performance metric results for training, testing, and tscv of all models (with FinBERT sentiment analysis variables).

Model	Set	R^2	MAE	MAPE	MSE	RMSE
FinBERT-LSTM	Training	0.9914	0.0092	0.0535	0.0002	0.0144
	Test	0.9808	0.0214	0.0373	0.0008	0.0286
	Time series CV (5-Fold)	0.9632	0.0128	0.0439	0.0004	0.0175
FinBERT-GRU	Training	0.9913	0.0120	0.0667	0.0002	0.0145
	Test	0.9509	0.0589	0.0506	0.0021	0.0458
	Time series CV (5-Fold)	0.9401	0.0243	0.0532	0.0007	0.0215
FinBERT-CNN	Training	0.9933	0.0088	0.0569	0.0002	0.0127
	Test	0.9452	0.0337	0.0528	0.0023	0.0484
	Time series CV (5-Fold)	0.9455	0.0151	0.0475	0.0007	0.0209
FinBERT-XGB	Training	0.9960	0.0062	0.0420	0.0001	0.0098
	Test	0.4933	0.0799	0.0989	0.0218	0.1477
	Time series CV (5-Fold)	0.8122	0.0243	0.0497	0.0053	0.0401
FinBERT-RF	Training	0.9973	0.0049	0.0261	0.0001	0.0080
	Test	0.0882	0.1348	0.1827	0.0389	0.1972
	Time series CV (5-Fold)	0.6776	0.0365	0.0618	0.0094	0.0501
FinBERT-LGBM	Training	0.9972	0.0052	0.0300	0.0001	0.0082
	Test	0.4481	0.0849	0.1068	0.0235	0.1534
	Time series CV (5-Fold)	0.7990	0.0247	0.0461	0.0057	0.0404
FinBERT-SVR	Training	0.9939	0.0094	0.0873	0.0001	0.0122
	Test	0.9787	0.0223	0.0343	0.0009	0.0303
	Time series CV (5-Fold)	0.9308	0.0126	0.0636	0.0003	0.0163

Note: The bold font has been used to highlight the model that shows the most successful prediction performance for each evaluation metric.

To quantify the added value of sentiment analysis variables extracted via the FinBERT model, we calculated the percentage change in prediction accuracy by comparing model performance before and after the integration of sentiment indicators. Table 10 presents the percentage improvements in MAE, RMSE, and R^2 for each model on the test set. Positive percentages in R^2 and negative percentages in error metrics (MAE and RMSE) indicate enhanced predictive performance. As shown, FinBERT-based models achieved significant performance gains, particularly in deep learning architectures.

Based on these results, the FinBERT-LSTM model is determined as the model that provides the lowest error rate and the highest prediction accuracy with the integration of news sentiment variables obtained from FinBERT sentiment analysis. These findings clearly demonstrate the importance of news sentiment variables for Bitcoin price prediction and the superiority of FinBERT-based DL models. The training and validation loss graph of the FinBERT-LSTM model is shown in Figure 7.

TABLE 10 | Percentage improvement after integrating FinBERT sentiment features into models (test set).

Model	Δ MAE (%)	Δ RMSE (%)	ΔR^2 (%)
LSTM	29.84%	34.85%	2.72%
GRU	0.0%	39.9%	10.07%
CNN	17.2%	4.91%	0.63%
XGB	−5.41%	−3.72%	−6.63%
RF	−12.61%	−7.7%	−58.79%
LGBM	1.62%	0.78%	1.86%
SVR	9.72%	7.9%	0.4%

When the loss graph is analyzed, the training loss (blue line) and validation loss (red line) of the model have decreased to very low levels and stabilized. In the first epochs (first few iterations), the validation loss decreases rapidly, while the training loss shows a similar downward trend. After about epoch 10, both training and validation loss reach minimal levels, stabilizing the model learning process. No overfitting problem is observed because the validation loss follows a parallel trend to the training loss and does not deviate excessively in the later stages of training. This finding shows that the FinBERT-LSTM model has a good generalization capacity and is optimized in a balanced way during training. In addition, the graph in Figure 8 shows the Bitcoin price prediction performance of the FinBERT-LSTM model on the test set.

The actual Bitcoin price (blue dots) and the price predicted by the model (red line) overlap to a large extent, indicating that the model has a successful forecasting capacity. The predicted values are highly correlated with the actual Bitcoin price. The model captured the direction of the market quite well, especially during large price spikes and drops. At the points of sudden price fluctuations (e.g., in early 2024 and mid-2024), the model accurately followed the overall trend, albeit with small deviations. The model's continued forecast accuracy, especially during volatile periods, suggests that the model's generalization capacity is strong. In conclusion, the FinBERT-LSTM model predicted the Bitcoin price quite successfully in the test set and captured the market trends with strong accuracy. Especially with the addition of news sentiment variables, the model's prediction accuracy improved and it adapted better to sudden changes in the market. The Feature Importance graph in Figure 9 shows which variables are most effective in the Bitcoin price prediction of the FinBERT-LSTM model.

The relative contribution levels of the features shown in Figure 9 allow us to analyze which factors the FinBERT-LSTM

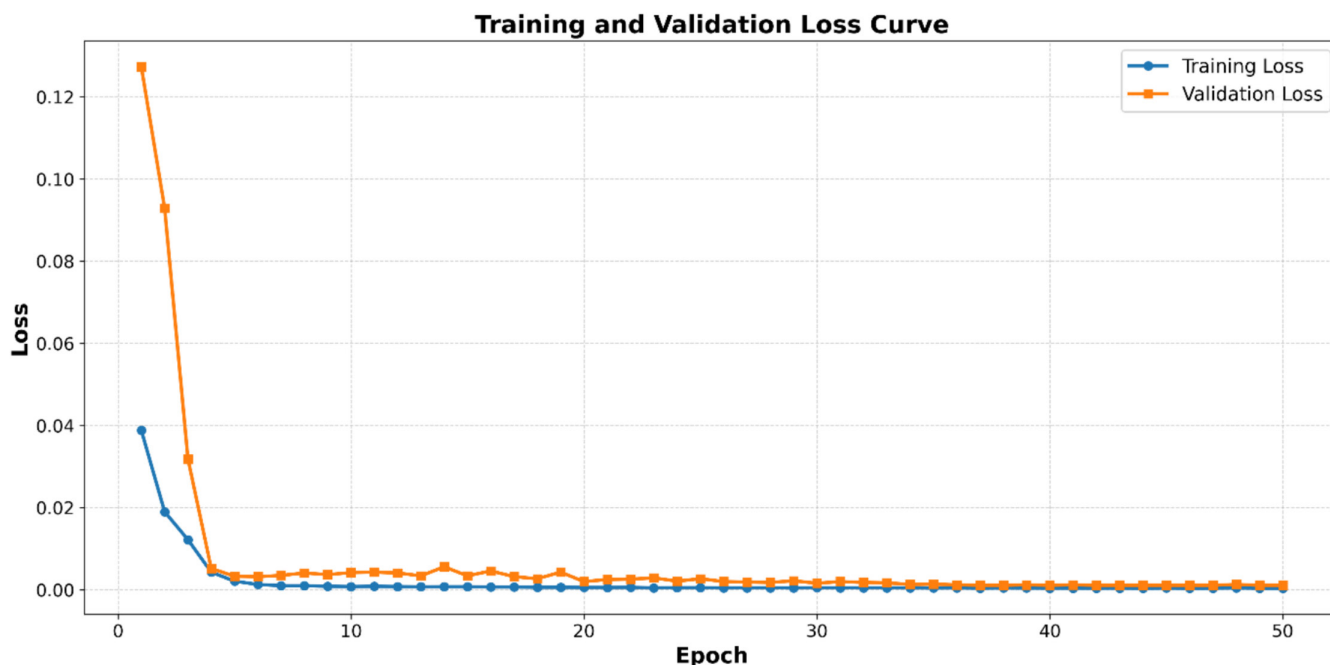


FIGURE 7 | Training and validation loss graph of FinBERT-LSTM model.

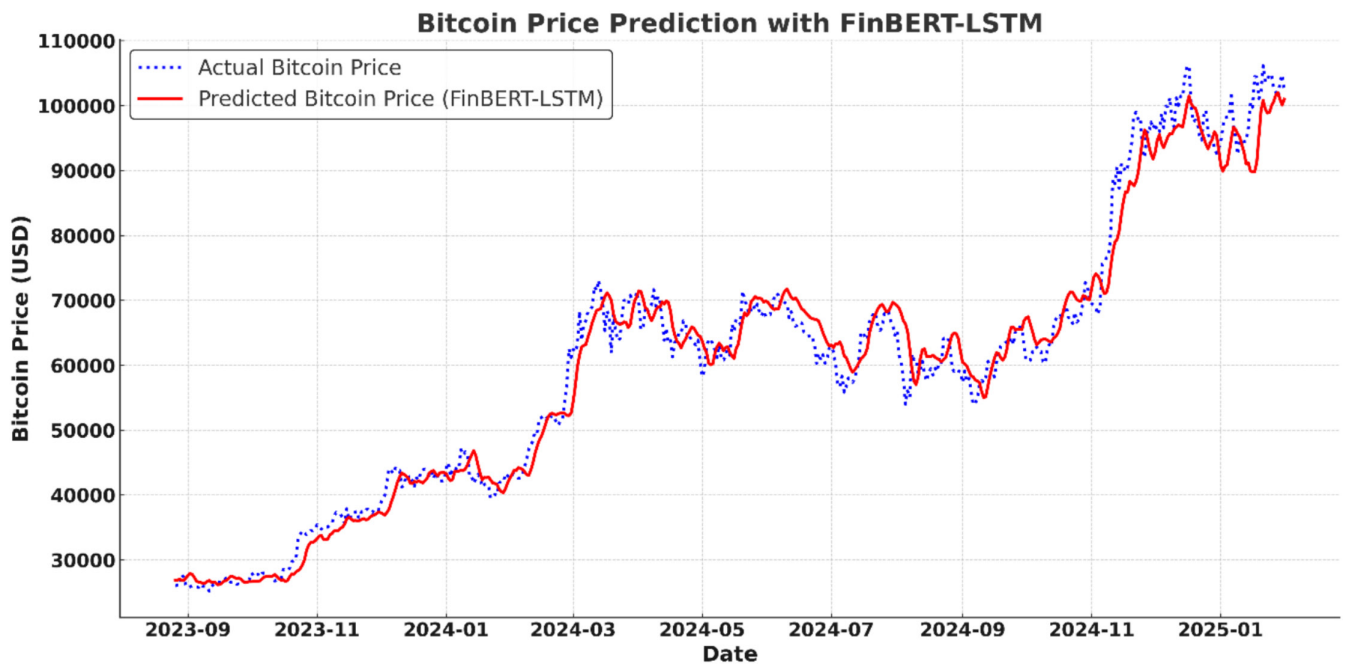


FIGURE 8 | Test prediction of the FinBERT-LSTM model.

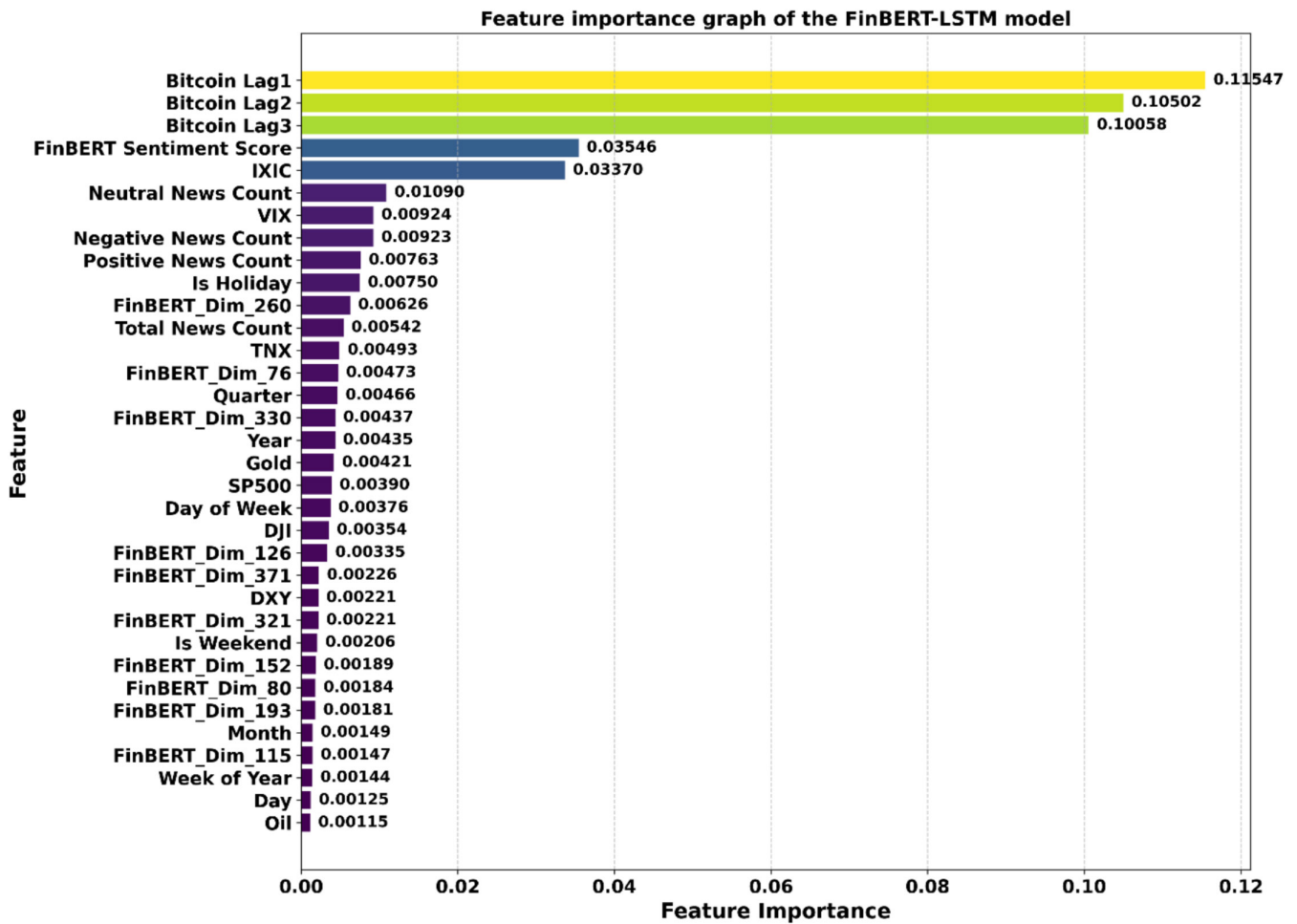


FIGURE 9 | Feature importance plot of the FinBERT-LSTM model.

model considers more important. Bitcoin Lag Values have the highest importance scores and contribute the most to the model's predictive capacity. This result shows that Bitcoin's past price movements play a critical role in predicting the future price. The "FinBERT Sentiment" variable (0.03546) achieved a very high importance score, confirming that the average daily sentiment scores of news are an important factor affecting the Bitcoin price. Among financial and macroeconomic indicators, the NASDAQ Composite Index (IXIC, 0.03370) and the VIX (0.00924) have relatively high importance scores. This suggests that US tech stocks influence the Bitcoin price along with the overall market risk appetite. Neutral, Negative and Positive News counts variables have significant significance scores for the model. This suggests that the number of positive, negative and neutral news items shared on a daily basis has an impact on price movements. Moreover, the best features extracted from FinBERT (e.g., FinBERT_Dim_260, FinBERT_Dim_76 and FinBERT_Dim_330) indicate that sentiment analysis vectors are found to be significant by the model. Other important factors include holidays (Is Holiday), the TNX, financial indicators such as S&P 500 and gold prices. However, Figure 10 shows how financial and macroeconomic indicators are evaluated by the model for Bitcoin price

prediction, along with their time-varying importance scores. Figure 11 shows the importance scores of positive, negative and neutral news over time. In these graphs, global macroeconomic events related to Bitcoin are shown at specific points. These events allow us to analyze how they shape Bitcoin price movements and the model's variable importance scores. Global events include Trump's crypto-friendly statements (October 2023), SEC Spot Bitcoin ETF approval (January 2024), Bitcoin Halving (March 2024), crypto sell-off and 25% drop (July 2024), US presidential election and Trump's re-election (November 2024), and Elon Musk's Blockchain announcement (January 2025).

According to the findings in Figure 10, IXIC has been one of the variables with the highest importance score, especially during events such as the SEC's Bitcoin ETF approval, the Bitcoin halving, and major selling waves. This shows that Bitcoin is increasingly integrated with traditional financial markets and major economic decisions have a direct impact on the crypto market. The VIX, on the other hand, increased during uncertain periods and was directly associated with market volatility. Moreover, political events, such as the US presidential elections, have also been observed to affect the

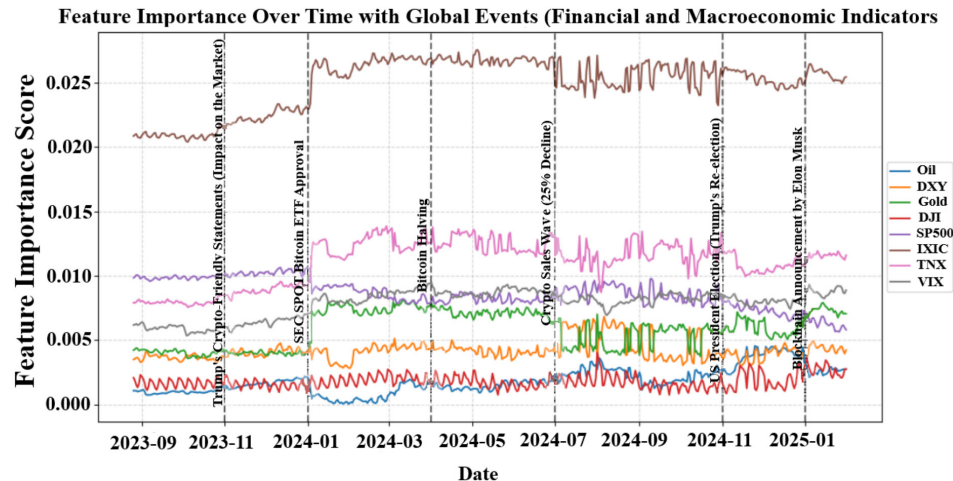


FIGURE 10 | Over time changing importance scores of financial and macroeconomic indicators.

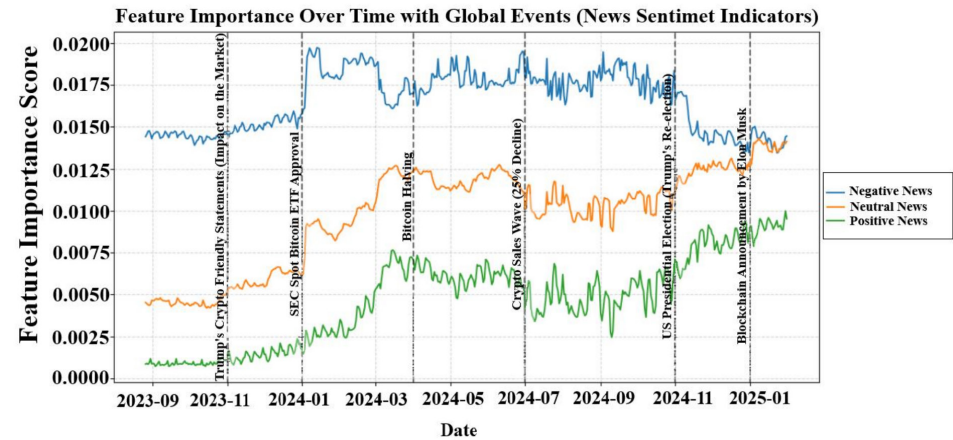


FIGURE 11 | Over time changing importance scores of positive, negative and neutral news.

Bitcoin price, with market uncertainty increasing especially during election periods. The relationship between traditional investment instruments such as the SP500, TNX and Gold and Bitcoin has fluctuated over time, with significant shifts in investor sentiment towards Bitcoin, especially during major macroeconomic events. On the other hand, according to the results in Figure 11, the importance scores of positive news increased significantly, especially during positive developments such as Trump's pro-cryptocurrency statements and the SEC's Bitcoin ETF approval, proving that investor sentiment has a direct impact on the Bitcoin price. Events such as the Bitcoin halving and major crypto sell-offs led to an increase in the importance scores of negative news, demonstrating that shifts in market sentiment have a strong impact on price dynamics. Neutral news, on the other hand, was generally more important during periods of market stabilization, suggesting that investors may focus more on neutral news during periods of uncertainty. Overall, these findings suggest that news-induced sentiment changes are a critical variable in forecasting the Bitcoin price and that integrating FinBERT-based sentiment analysis into forecasting models significantly improves the performance of the model. In this context, the proposed hybrid model, which incorporates both macroeconomic indicators and news-based sentiment analysis, provides more accurate forecasts of Bitcoin market volatility and outperforms traditional forecasting approaches.

4.2 | What-If Analysis

The Impact of Different Market and Sentiment Scenarios on the Bitcoin Price.

What-if analysis was conducted to evaluate the impact of different market and sentiment scenarios on the Bitcoin price. Within the scope of this analysis, different sentiment scenarios are applied to the FinBERT-LSTM model and the forecasts of the model are compared. First, we evaluate the impact of news sentiment variables on the model's forecasting performance, and then we analyze how price forecasts change under different sentiment scenarios. In addition, the contribution of the news embedding vectors obtained with the FinBERT model to the forecast accuracy is analyzed and how these features improve the overall performance of the model is detailed. Within the scope of these analyses, the impact of changes in the number of negative and positive news on the Bitcoin price, the role of news sentiment on the error metrics in model forecasts, and the contribution of semantic information from news to price forecasts are discussed in detail. As a result, the impact of news sentiment on Bitcoin price forecasting models is visualized and the potential of the FinBERT model in financial forecasting processes is demonstrated. The graph presented in Figure 12 shows how the model responds to percentage increases in the number of negative and positive news in Bitcoin price forecasts. To operationalize these scenarios, we introduced incremental sentiment shocks of $\pm 10\%$, $\pm 20\%$, $\pm 30\%$, $\pm 40\%$, and $\pm 50\%$ to the FinBERT-derived sentiment variables (positive, negative, and neutral scores). These levels were designed to represent both routine and extreme market conditions. Lower thresholds ($\pm 10\%$ – 20%) capture

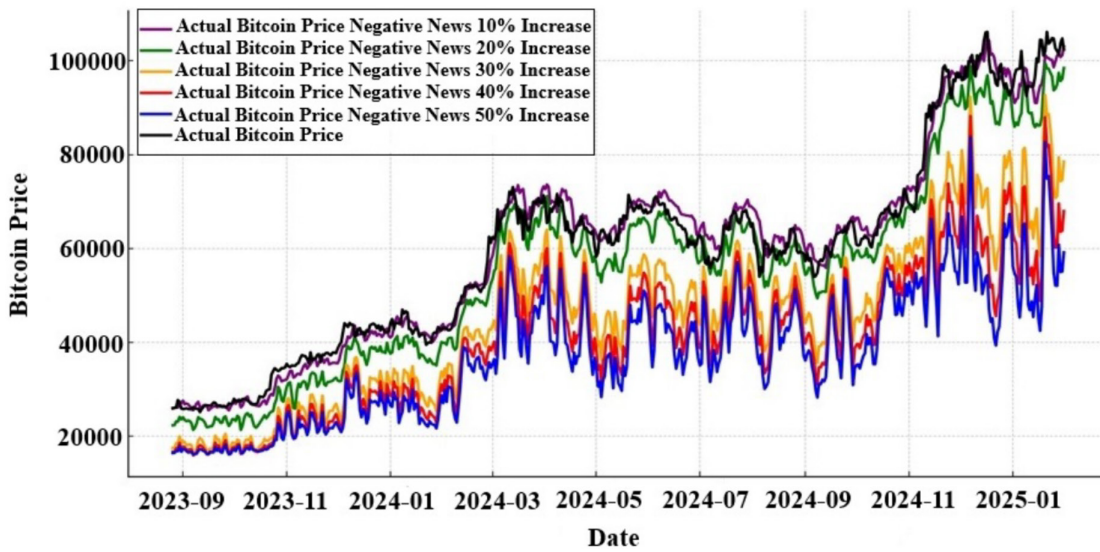
the effects of typical daily news variations, while larger shocks ($\pm 30\%$ – 50%) simulate extraordinary stress events such as regulatory interventions, exchange hacks, or global uncertainty episodes frequently observed in cryptocurrency markets. Similar approaches have been employed in recent empirical research, which demonstrates that even moderate increases in negative sentiment can lead to substantial volatility spikes (Aslanidis et al. 2024), while extreme sentiment shocks amplify downturns and clustering effects (Sakariyahu et al. 2024). Moreover, evidence from social-media-driven sentiment shows that negative tones produce immediate and asymmetric volatility responses compared to positive tones (Lupu and Donoiu 2025). Incorporating this design ensures that our scenario analysis is aligned with the latest literature and provides a realistic test of how varying levels of sentiment shocks influence Bitcoin price dynamics.

In scenarios where the number of negative news stories is increased by 10%–50%, a significant decrease in Bitcoin price forecasts is observed. The severity of this effect grows in parallel with the percentage increase in negative news. In particular, at negative news increases of 40% and 50%, the estimated Bitcoin price declines significantly. In contrast, the impact of an increase in the number of positive news on the model's predictions was relatively limited. While there is an upward trend in prices as positive news increases, this effect is weaker than the effect of negative news. This supports the hypothesis that cryptocurrency markets are more sensitive to negative news. Figure 13 evaluates the effect of the FinBERT sentiment variable on Bitcoin price forecasts in different scenarios (when sentiment is -1 , 0 , and 1 on all dates, respectively).

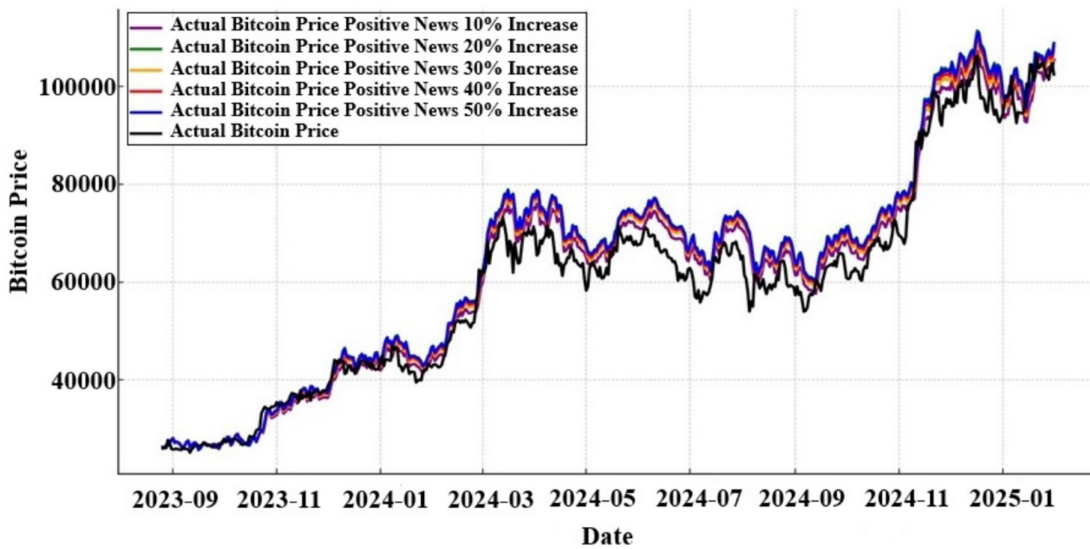
The sentiment scores obtained with the FinBERT model are decomposed into negative, neutral and positive sentiment scores for the entire test set period. The results show that negative sentiment news flows have a significant downward impact on Bitcoin price forecasts, while neutral news flows have a moderate impact on prices, and news with positive sentiment supports prices upwards. It is noteworthy that the impact of negative news is much more pronounced than that of other categories. In addition, Figure 14 evaluates the percentage change in Bitcoin price forecasts for these scenarios.

When the “FinBERT Sentiment” variable is negative (-1) for all periods in the test set, the model predicts a 26.23% decline in the Bitcoin price. When the sentiment score is neutral (0), the drop in the Bitcoin price remains at 10.58%. When the sentiment score is positive ($+1$), there is a 7.27% increase in the predicted price. These results suggest that the market is highly reactive to news sentiment and that negative news flows depress price forecasts downwards. In addition, Figure 15 shows the effect of integrating the “FinBERT Sentiment” variable into the model on forecast accuracy through MAPE and R^2 scores. When the “FinBERT Sentiment” variable is added, the MAPE value improves by 6.54% and the R^2 score increases by 0.0781 points. This finding proves that integrating this variable into the model significantly improves the prediction accuracy.

However, the graph in Figure 16 analyzes the contribution of the integration of the embedding vectors obtained with the



(a) Bitcoin predictions of the model after percentage increase in the number of negative news



(b) Bitcoin predictions of the model after the percentage increase in the number of positive news

FIGURE 12 | Bitcoin predictions of the model after the percentage increase in the number of positive and negative news.

FinBERT model to the prediction performance. The MAPE value decreased from 5.69% before the inclusion of FinBERT embedding vectors to 3.73% after their inclusion. At the same time, the R^2 score increased by 0.0409 points. In terms of its impact on Bitcoin price forecasts, the model with FinBERT embedded vectors was found to be able to capture price changes more accurately and produce more accurate forecasts.

Taken as a whole, these findings suggest that news sentiment and semantic information derived from news content play a critical role in improving the accuracy of Bitcoin price prediction models. Negative news flows have a much higher downward impact on prices than positive news, and the integration of sentiment variables and embedded vectors obtained with the FinBERT model enables more accurate prediction of Bitcoin price movements.

4.3 | The Effect of GARCH Model and News Sentiment on Bitcoin Price Volatility

Modeling volatility in financial markets is of great importance for risk management and determining investment strategies. One of the most widely used models to predict the volatility of highly volatile crypto assets such as Bitcoin is the GARCH model. The GARCH(1, 1) model was developed to capture volatile volatility in financial time series and to model the dependence of conditional variance over time. In this study, it is applied to determine Bitcoin price volatility and to measure the impact of news sentiment on volatility. The GARCH(1, 1) model is one of the most widely used models to capture the volatility dynamics of financial markets (Khan et al. 2023; Marisetty 2024). This model estimates current volatility with past error terms (ARCH component) and past variance values (GARCH component).

What-If Analysis: Impact of FinBERT Sentiment on Bitcoin Price Prediction

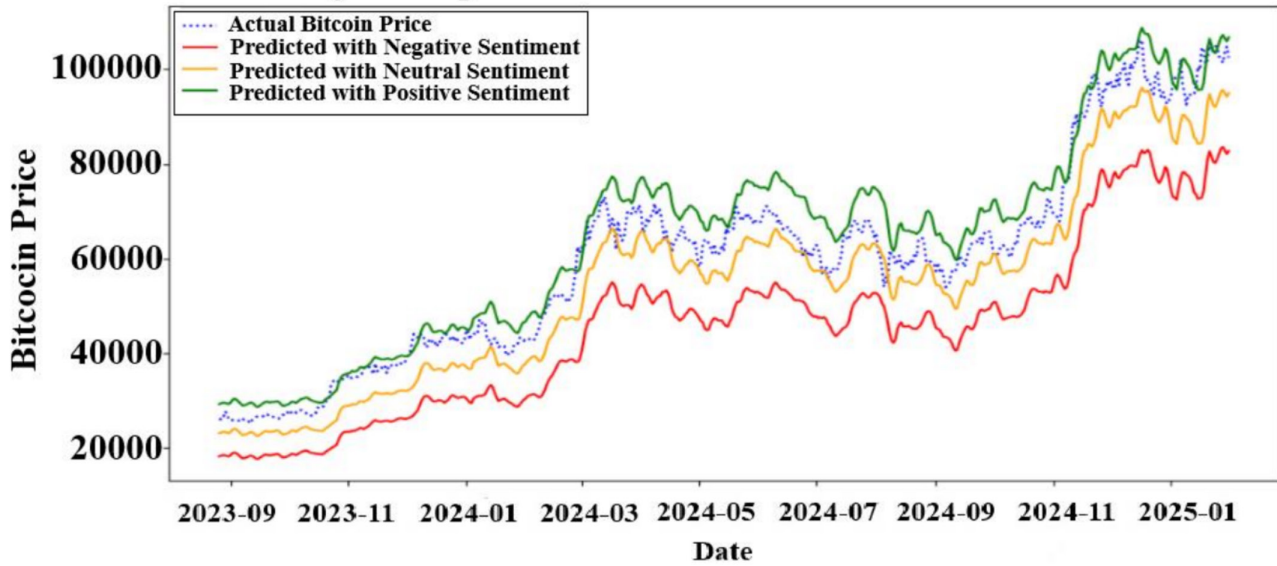


FIGURE 13 | The effect of sentiment variable on Bitcoin price forecasts in different scenarios.

Impact of Different FinBERT Sentiment Values on Bitcoin Price Predictions

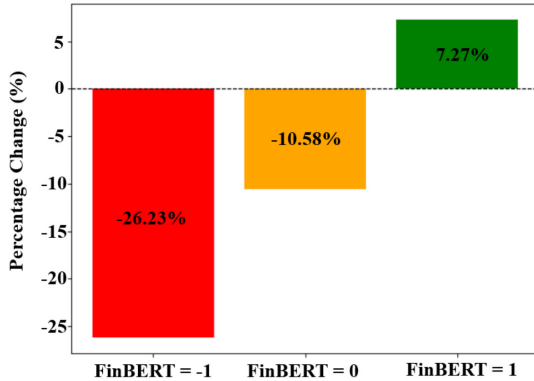


FIGURE 14 | Percentage effect of sentiment variable on Bitcoin price predictions in different scenarios.

The GARCH model allows the conditional variance to be modeled by past shocks and past variance values. The GARCH(1, 1) model is defined as in Equations (8) and (9).

$$r_t = \mu + \epsilon_t, \quad \epsilon_t = \sigma_t z_t \quad (8)$$

$$\sigma_t^2 = a_0 + a_1 \epsilon_{t-1}^2 + B_1 \sigma_{t-1}^2 \quad (9)$$

where r_t is the Bitcoin price return, μ is the average return, ϵ_t is the error term (white noise), σ_t^2 is the conditional variance (volatility), a_0 is the constant term, a_1 is the ARCH component (the effect of past error squares on volatility), B_1 is the GARCH component (the effect of past volatility on current volatility), and z_t is the stochastic process with a standard normal distribution.

In the GARCH(1, 1) model, when $\alpha_1 + \beta_1 < 1$ condition is satisfied, the model is considered to be stationary. When α_1 is high, the impact of new shocks on volatility is stronger, while when β_1 is high, volatility persists for a long time. In order to model the volatility of the Bitcoin price, we first calculate the daily logarithmic returns of the price series. The stationarity analysis

of the return series is evaluated with the Augmented Dickey-Fuller (ADF) test and the series is found to be stationary. Then, Ljung-Box and ARCH-LM tests are used to test for the presence of autocorrelation and heteroskedasticity in the series and the GARCH model is deemed appropriate for modeling volatility. These results are presented in Table 11.

According to the ADF test results, the ADF statistic is -24.226 , and the p -value is very close to 0. Since this value is lower than the 1% (0.01), 5% (0.05), and 10% (0.10) critical values, we conclude that the series is stationary even at the 1% significance level. In other words, the log returns series is stationary at the 99% confidence level. On the other hand, since the p -value of the Ljung-Box test is less than 0.05, we can conclude that there is a significant autocorrelation in the series. This supports the suitability of the GARCH model for volatility modeling. In addition, since the p -value of the ARCH-LM test is less than 0.05, we can conclude that there is significant heteroskedasticity in the series. This result suggests that volatility is volatile and the GARCH model may be appropriate. The GARCH(1, 1) model on the logarithmic returns of Bitcoin is applied to analyze the change in volatility over time and to measure the effect of news sentiment on volatility. Table 12 presents the main parameters and statistical results of the model. According to these results, volatility in the Bitcoin market tends to persist for a while (high beta value). Sudden price changes in the past may have a short-term impact, but the main determinant of volatility is past volatility. The GARCH model seems appropriate for modeling Bitcoin volatility.

The graph presented in Figure 17 shows the historical volatility of the Bitcoin price from a time-series perspective. The volatility dynamics estimated using the GARCH(1, 1) model reflect the transitions between periods of high volatility and phases of low volatility that financial markets typically exhibit. The chart shows that there are sharp increases in volatility during certain periods. These fluctuations may be due to exogenous factors such as macroeconomic factors, regulatory changes and large

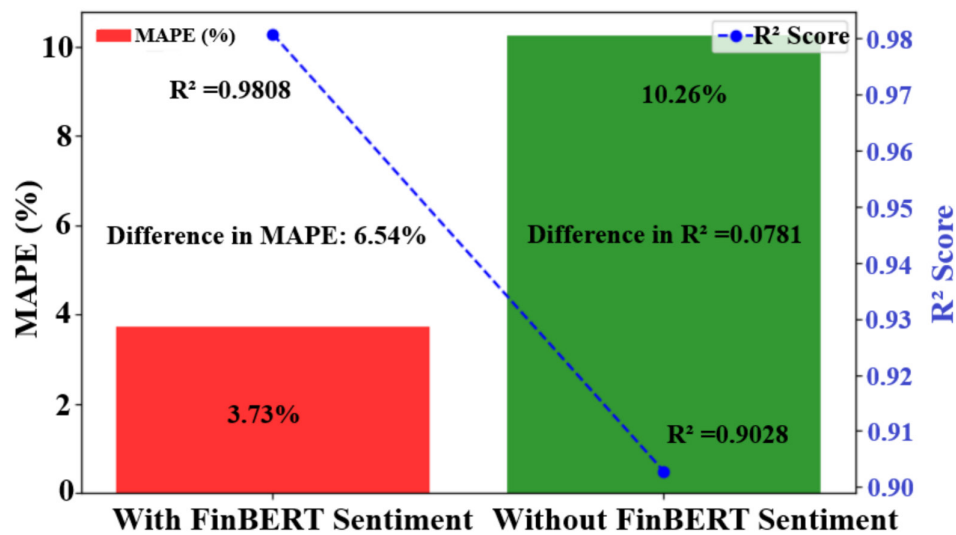


FIGURE 15 | Comparison of prediction errors with and without FinBERT sentiment variable.

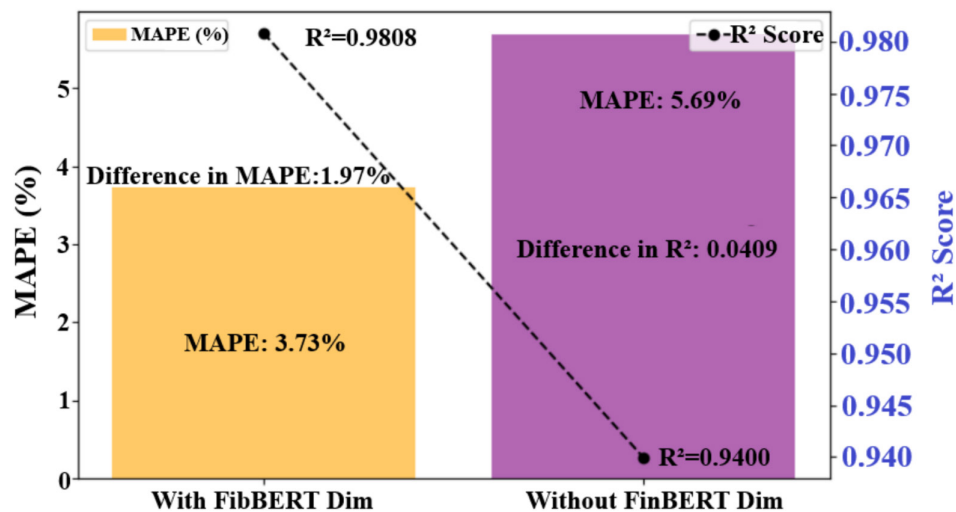


FIGURE 16 | Comparison of prediction errors with and without embedding vectors obtained with the FinBERT model.

investor movements. However, volatility appears to increase significantly in certain periods and then stabilize at a relatively lower level, suggesting that the Bitcoin market has a tendency to volatility clustering.

In addition, scenarios of 10%, 20%, 30%, 40%, and 50% increase in the number of news stories were created to determine the impact of increases in the number of positive and negative news stories on Bitcoin volatility and to reveal the severity of this effect. The results of this analysis are shown in Figure 18.

The findings in Figure 18 suggest that an increase in positive news leads to a downward trend in volatility. In particular, volatility decreases significantly in the 40% and 50% positive news increase scenarios. This result suggests that the market reacts more consistently to positive news flow and volatility decreases. On the other hand, an increase in negative news significantly increases price volatility. In particular, negative news spikes of 30% or more sharply increase volatility and increase market

TABLE 11 | ADF, Ljung-Box, and ARCH-LM results.

Log returns series	Value
ADF statistics	−24.2259
p	0.0000
Critical values (1%)	−3.4429
Critical values (5%)	−2.8671
Critical values (10%)	−2.5697
Ljung-Box statistic (10 lags)	25.4321
p	0.0123
ARCH-LM statistic	35.6789
p	0.0045
F statistic	4.3210
p	0.0032

uncertainty. This suggests that the market is highly sensitive to negative news and investors increase their risk premium during uncertain periods, leading to higher volatility. These results

TABLE 12 | GARCH(1, 1) model summary.

Parameter	Coefficient (Coef)	p
Mu (μ)—average model	2.6739e−03	0.0128
Omega (ω)—volatility model	1.0806e−04	0.0012
Alpha[1]—ARCH coefficient	0.0784	0.0611
Beta[1]—GARCH coefficient	0.7632	<0.0001
R-square	0.000	—
Adj. R-square	0.000	—
Log-likelihood	1174.36	—
AIC	−2340.72	—
BIC	−2323.68	—
Number of observations	524	—

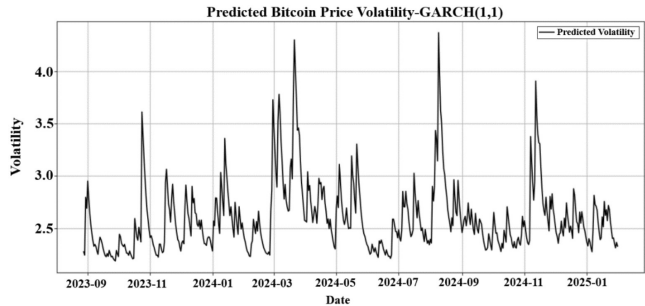
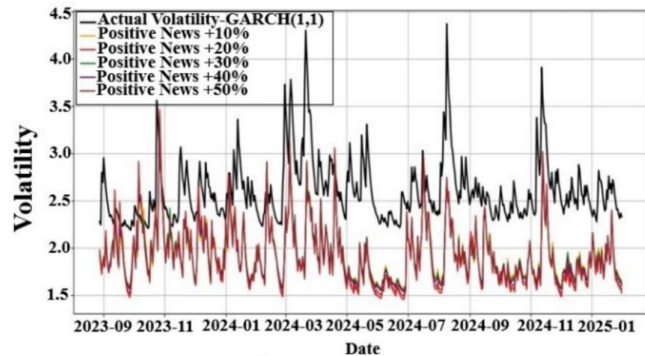


FIGURE 17 | Volatility dynamics estimated using the GARCH(1, 1) model.

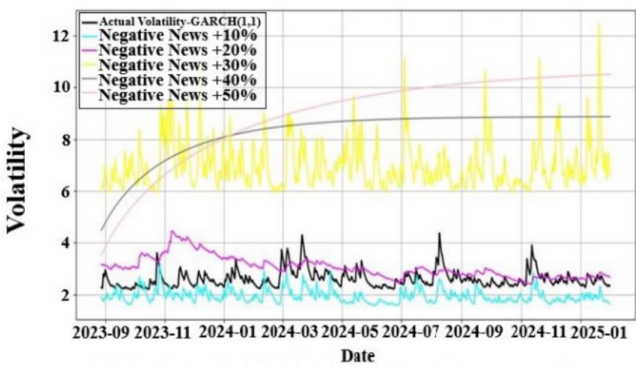


(a) Estimated volatility after the percentage increase in the number of positive news

suggest that the market is highly sensitive to news flow and sentiment analysis-based variables in financial models can play a critical role in volatility forecasting. They also suggest that investors and portfolio managers should shape their risk management strategies by taking news sentiment indicators into account.

5 | Discussion

In this study, we evaluate the effectiveness of hybrid models that combine macroeconomic factors, financial indicators and NLP-based variables such as news sentiment in predicting the Bitcoin price accurately and reliably. Our findings show that the volatile and complex nature of the Bitcoin price, which is frequently emphasized in the literature (Noviandy et al. 2023; Jesika et al. 2023), can only be better explained by comprehensive hybrid models. According to the results, the addition of FinBERT-based sentiment analysis results and embedding vectors significantly improved the overall performance of the forecasting models. The FinBERT-LSTM model outperformed both ML and other DL models with its high accuracy ($R^2=0.9808$; MAPE = 3.73%), especially on the test dataset. Although classical machine learning methods such as SVR have been observed to provide successful results in Bitcoin price prediction in the literature (Khedr et al. 2021), the superiority of the DL-based FinBERT-LSTM model has been clearly demonstrated after FinBERT integration. This shows that DL models are more effective in capturing complex and nonlinear relationships and successfully model the interaction of news sentiment and financial market dynamics. The findings, supported by what-if analyses, also show that negative news has a higher potential to affect the Bitcoin price downwards (up to 50% increase scenario) than positive news has a higher potential to affect the price upwards. This is consistent with similar studies in the literature (Mgadmi 2023; Jung et al. 2023). Moreover, volatility analysis using the GARCH(1, 1) model shows that an increase in negative news has a strong increasing effect on volatility, while positive news decreases volatility. The use of the GARCH model is consistent with the persistence property of Bitcoin's volatility and is in line with the results of similar studies in the literature (Sahi et al. 2023; Zhang et al. 2024).



(b) Estimated volatility after the percentage increase in the number of negative news

FIGURE 18 | The impact of 10%, 20%, 30%, 30%, 40%, and 50% increases in the number of news articles on Bitcoin volatility.

Building on these findings, it is important to clarify how the what-if shocks were designed and how their outcomes should be interpreted. In our scenario experiments, shocks of $\pm 10\%$, $\pm 20\%$, $\pm 30\%$, $\pm 40\%$, and $\pm 50\%$ were applied to FinBERT-derived sentiment variables. Lower-level shocks ($\pm 10\%$ – 20%) were intended to capture typical daily fluctuations in market sentiment, while higher-level shocks ($\pm 30\%$ – 50%) simulated more extreme stress conditions such as regulatory announcements, exchange hacks, or global uncertainty episodes. This design reflects the high-volatility environment of cryptocurrency markets and aligns with recent empirical research: Aslanidis et al. (2024) show that even moderate increases in negative sentiment strongly correlate with downward price pressure, while Sakariyahu et al. (2024) highlight that negative sentiment coupled with global uncertainty exacerbates volatility clustering. Likewise, Lupu and Donoiu (2025) find that sharp increases in negative sentiment extracted from social media produce immediate volatility spikes, whereas positive shocks lead to weaker and delayed price adjustments.

To further contextualize these results, it is important to emphasize their economic and behavioral implications. In highly speculative and volatile markets such as cryptocurrency, negative shocks—particularly those reflected in news sentiment—tend to trigger stronger risk-averse and loss-averse behaviors than positive shocks encourage optimism. Recent empirical evidence supports this interpretation: Long et al. (2024) find that social media sentiment exhibits asymmetric effects on cryptocurrency markets, with heightened sensitivity during negative sentiment periods. Additionally, Lupu and Donoiu (2025) demonstrate that negative sentiment extracted from Twitter leads to immediate spikes in volatility and trading volume, whereas positive sentiment effects are more muted and delayed. These results imply that the asymmetric impact we observe is not only statistical but also rooted in investor psychology and market structure. Economically, this suggests that for investors and risk managers, negative sentiment flows may serve as earlier and stronger warning signals of downward price movements and volatility spikes than positive sentiment can provide for upward corrections.

In conclusion, it should be emphasized that the developed hybrid approach provides a powerful and innovative modeling framework for Bitcoin price forecasting beyond existing models. Moreover, the asymmetric sentiment effects we observe have both theoretical and practical importance: by underscoring how investor sentiment—especially negative sentiment—can shape market dynamics more violently, our results suggest that forecasting models and risk management strategies in the cryptocurrency space need to pay special attention to adverse news flows.

6 | Conclusion

This study shows that the FinBERT-LSTM model, which uses financial and macroeconomic variables (e.g., DXY, VIX, TNX, IXIC), temporal factors (e.g., quarter, month, weekend), financial indicators (Bitcoin lagged variables), and sentiment scores and embedding vectors derived from news headlines in a hybrid format, outperforms in Bitcoin price forecasting. The study

demonstrates that combining news-derived sentiment analysis features with financial and macroeconomic indicators strengthens Bitcoin price forecasting. Incorporating FinBERT sentiment scores and embedding vectors leads to a clear improvement in the predictive performance of both DL and ML models. The results also show that the FinBERT-LSTM model provides the highest level of forecasting accuracy. In addition, the analysis of positive, negative, and neutral FinBERT news within the GARCH volatility framework indicates that negative news has a considerably stronger influence on Bitcoin price volatility compared to positive news.

The innovative aspect of this study is that it integrates macroeconomic, financial and sentiment data, which are usually treated separately in the existing literature, into a single hybrid model. This integration yields better results in Bitcoin price forecasting than existing models and clearly demonstrates the critical impact of news sentiment on market volatility. Moreover, the study contributes to the literature by demonstrating the ability of DL (especially LSTM) models to handle such hybrid data sets.

The first limitation of this study is that it focuses only on Bitcoin. The generalizability of the results can be increased by analyzing other cryptocurrencies other than Bitcoin with similar methods. The second limitation is that the news data was obtained only from certain sources (Reuters, Bloomberg, Investing, CoinDesk, CryptoSlate, Google News, and GDELT); the inclusion of different social media platforms and larger data sets may increase the generalizability of the results. As a third limitation, the hyperparameter optimization processes of the models used are based on a limited number of experiments. In future research, higher performances can be achieved by applying more comprehensive optimization strategies. In future studies, it is suggested to extend similar analyses to different crypto assets such as Ethereum, Ripple, and so forth. Furthermore, the use of real-time sentiment data and the evaluation of more recent Transformer-based NLP models (such as the GPT series) can further enrich the prediction accuracy and market analysis. Finally, additional analysis is recommended to assess the effects of news sentiment on different investment strategies. These analyses will provide more practically useful results for market practitioners.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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